Highlights

Elite-Guided Dynamic Feature Probability Sparrow Search Algorithm for High-Dimensional Feature Selection

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- Dynamic feature probability with Bernoulli sampling bypasses the 0.5 bottleneck.
- Elite consensus tuning with delayed activation corrects probabilities by voting.
- A time-varying producer ratio smoothly shifts the exploration-exploitation balance.
- EDP-SSA attains 91.88% accuracy selecting only 1.04% features on 16 datasets.
- Wilcoxon, Friedman and ablation studies confirm significance of components.

Elite-Guided Dynamic Feature Probability Sparrow Search Algorithm for High-Dimensional Feature Selection

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ARTICLE INFO

Keywords:
Feature selection
Sparrow search algorithm
Dynamic feature probability
Elite consensus
High-dimensional microarray data
Swarm intelligence

ABSTRACT

High-dimensional microarray data, characterized by thousands of features but only tens or hundreds of samples, impose severe computational and generalization burdens on classification models. Feature selection has become an essential preprocessing step to build accurate and interpretable classifiers from such data. Among metaheuristic approaches, the Sparrow Search Algorithm (SSA) has shown promise for wrapper-based feature selection, yet existing binary SSA variants rely on transfer functions that create an inherent probability bottleneck near 0.5, making it difficult to decisively exclude irrelevant features from high-dimensional spaces. This paper proposes an Elite-Guided Dynamic Feature Probability Sparrow Search Algorithm (EDP-SSA) that replaces the conventional transfer-function binarization with a dynamic feature probability (DFP) vector directly updated through Bernoulli sampling. The DFP vector is initialized at a low value (0.1) and adjusted at each iteration based on the observed selection frequency across the population, implementing a frequency-driven elimination mechanism. An elite consensus tuning (ECT) strategy with delayed activation maintains a sliding window of the best binary masks from recent generations and applies majority voting to reinforce features consistently selected by high-quality solutions. A time-varying adaptive producer ratio increases linearly from 15% to 35% during evolution to balance exploration and exploitation, and a greedy single-feature flipping local search refines the final subset under a bounded evaluation budget. Experiments on 16 high-dimensional microarray datasets (2,000–22,283 features) against 10 competing algorithms demonstrate that EDP-SSA achieves the highest average classification accuracy of 91.88% while selecting only 1.04% of features on average, compared with 89.27% accuracy and 7.98% features for the best competitor (bIRIME). Wilcoxon signed-rank tests at the 0.05 significance level confirm statistically significant improvements in 140 of 160 pairwise comparisons, and the Friedman test ranks EDP-SSA first with an average rank of 1.125. Ablation experiments quantify the progressive contribution of each component: the DFP mechanism raises the average accuracy from 85.30% (basic SSA) to 88.03%, the ECT provides an additional improvement to 89.19%, and the full model reaches 91.88%.

1. Introduction

The rapid advancement of high-throughput technologies has led to an increasing number of applications characterized by high-dimensional data, including genomics, proteomics, medical image analysis, and text classification. Such datasets typically contain thousands or even tens of thousands of features, while the number of available samples remains relatively small. This "large p, small n" characteristic poses significant challenges for machine learning algorithms, including increased computational cost, overfitting risk, and degraded generalization performance [14, 23]. Feature selection, which aims to identify the most discriminative subset of features while removing redundant and irrelevant ones, has therefore become an indispensable preprocessing step in high-dimensional data analysis.

Feature selection methods are generally categorized into three groups: filter, wrapper, and embedded approaches. Filter methods evaluate features based on statistical measures such as mutual information, correlation, or the t-test, independently of any classifier. Although computationally efficient, they typically ignore feature interactions and may miss complementary feature combinations. Wrapper methods employ a specific classifier to evaluate candidate feature subsets and generally achieve higher classification accuracy at the cost of greater computational demand. Embedded methods integrate feature selection into the model training process, as exemplified by LASSO and decision-tree-based approaches. Among these, wrapper methods combined with metaheuristic optimization have attracted considerable attention owing to their flexibility and strong search capabilities.

Metaheuristic algorithms, particularly swarm intelligence methods, have been extensively applied to wrapper-based feature selection. Representative algorithms include genetic algorithms (GA), particle swarm optimization (PSO), ant colony optimization (ACO), grey wolf optimizer (GWO), and the sparrow search algorithm (SSA). The SSA, proposed by Xue and Shen [42], is a nature-inspired optimization algorithm that simulates the foraging and anti-predation behaviors of sparrows. It has gained popularity in feature selection due to its fast convergence, few control parameters, and ease of implementation. Several binary variants of SSA have been developed for feature selection, typically employing transfer functions—such as sigmoid, V-shaped, or U-shaped functions—to map continuous positions to binary solutions.

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Despite these efforts, existing binary SSA variants suffer from three fundamental limitations when applied to high-dimensional feature selection. First, the dependence on transfer functions creates an inherent probability bottleneck: selection probabilities tend to cluster around 0.5, making it difficult to decisively exclude irrelevant features from high-dimensional spaces. Second, the producer–scrounger ratio in standard SSA remains fixed throughout the search, which prevents a dynamic balance between global exploration and local exploitation. Third, the scrounger update mechanism relies largely on random perturbations, which limits the algorithm's ability to refine solutions in later evolutionary stages.

To overcome these limitations, this paper proposes an Elite-Guided Dynamic Feature Probability Sparrow Search Algorithm (EDP-SSA) for high-dimensional feature selection. The core idea is to replace the conventional transfer-function-based binarization with a dynamic feature probability (DFP) mechanism that directly maintains and evolves a probability vector over the feature space. Specifically, the DFP vector is initialized at a low value (0.1) and updated at each iteration based on the observed selection frequency of each feature across the population. Features that are frequently selected (above a high-frequency threshold) have their probabilities decreased, while rarely selected features (below a low-frequency threshold) also have their probabilities reduced, thereby implementing a frequency-based elimination strategy. Binary masks are generated by Bernoulli sampling from the DFP vector, enabling the algorithm to naturally converge toward compact feature subsets without being constrained by any transfer function.

To further strengthen the search guidance, an elite consensus tuning (ECT) mechanism is introduced. A sliding window maintains the best binary masks from the most recent generations. After a predefined delay period, majority voting across the elite pool identifies features that are consistently selected by high-quality solutions, and their probabilities are reinforced. Conversely, features that are unanimously excluded across all elite solutions have their probabilities further suppressed. The delayed activation ensures sufficient exploration in the early stage before the elite information is trusted. A time-varying adaptive producer ratio increases linearly from 15% to 35% over the course of evolution, enabling broader exploration initially and progressively more exploitation. Finally, a greedy single-feature flipping local search is applied to the best solution after the evolutionary process, subject to a bounded evaluation budget, to further refine the selected subset.

The proposed algorithm is evaluated on 16 high-dimensional microarray datasets, with feature dimensions ranging from 2,000 to 22,283, and compared against 10 well-established feature selection algorithms, including three SSA-based methods (SSA, iBSSA, and ESSSO) and seven other metaheuristic or filter approaches (BPSO, BGWO, BPSO-V, BinCOA, bIRIME, BMPA-TVSinV, and SVM-RFE). All algorithms are evaluated under identical experimental conditions: a population size of 100, 100 iterations, KNN with $k = 5$ as the base classifier, 5-fold cross-validation during the search phase, and 10-fold cross-validation for final performance assessment. Each algorithm is executed with 10 different random seeds to ensure statistical reliability.

EDP-SSA achieves the highest average classification accuracy of 91.88% across all 16 datasets, outperforming the second-best method (bIRIME, 89.27%) by 2.61 percentage points. More importantly, EDP-SSA selects only 1.04% of features on average, compared to 7.98% for the nearest competitor, confirming its superior feature compression capability. The algorithm also obtains the best average fitness value of 0.0436, substantially lower than competing methods. Wilcoxon signed-rank tests at the 0.05 significance level confirm statistically significant improvements in 140 of the 160 pairwise comparisons, with only one significant loss (against bIRIME on lung). Ablation experiments further verify that each proposed mechanism contributes indispensably, with the DFP mechanism raising the average accuracy from 85.30% for basic SSA to 88.03%, the elite consensus tuning providing an additional improvement to 89.19%, and the full model reaching 91.88%.

The main contributions of this paper are summarized as follows:

1. A dynamic feature probability mechanism that directly evolves a probability vector via Bernoulli sampling, breaking through the 0.5 bottleneck imposed by conventional transfer functions and enabling effective feature elimination in high-dimensional spaces.
2. An elite consensus tuning strategy with delayed activation that exploits historical best solutions through majority voting, providing robust macro-level probability correction without premature convergence.
3. A time-varying adaptive producer ratio that smoothly transitions the search balance from exploration to exploitation, complemented by a greedy local search post-processing step.
4. Evaluation on 16 high-dimensional microarray datasets against 10 competing algorithms, with ablation studies and parameter sensitivity analysis.

The remainder of this paper is organized as follows. Section 2 reviews related work on feature selection methods and swarm intelligence approaches. Section 3 presents the proposed EDP-SSA algorithm in detail. Section 4 describes the experimental setup and discusses the results. Section 5 concludes the paper and outlines future research directions.

2. Related Work

2.1. Feature Selection Methods

Feature selection aims to identify a compact and discriminative subset of features from the original high-dimensional space. Classical filter methods, such as mutual information (MI) [30], chi-square test [23], and Relief [21], evaluate each feature independently based on its statistical relationship with the class labels. While computationally efficient, filter

methods generally ignore feature–feature interactions and may fail to identify complementary feature combinations that are jointly informative but individually weak.

Wrapper methods address this limitation by employing a learning algorithm to evaluate candidate feature subsets. Common search strategies include forward selection, backward elimination, and recursive feature elimination (RFE) [22]. Support vector machine recursive feature elimination (SVM-RFE) [16] has been widely used in bioinformatics for gene selection from microarray data. Although wrapper methods typically achieve higher classification accuracy than filter methods, their computational cost grows exponentially with the number of features, making them less practical for very high-dimensional problems without efficient search strategies.

Embedded methods, such as LASSO [38], elastic net [43], and tree-based feature importance [4], perform feature selection as part of the model training process. These methods offer a good balance between efficiency and accuracy but may be limited by the specific model assumptions and may not generalize well across different classification tasks.

2.2. Swarm Intelligence for Feature Selection

In recent years, swarm intelligence algorithms have emerged as a popular choice for wrapper-based feature selection due to their ability to explore large combinatorial search spaces efficiently. Genetic algorithms (GA) [17] use selection, crossover, and mutation operators to evolve a population of feature subsets. Binary particle swarm optimization (BPSO) [19] adapts the continuous PSO framework to binary spaces using transfer functions. Binary grey wolf optimizer (BGWO) [8] and binary ant colony optimization (ACO) [7] have also been applied to feature selection with promising results.

The sparrow search algorithm (SSA), proposed by Xue and Shen [42], is a relatively recent swarm intelligence method inspired by the foraging and anti-predation behaviors of sparrows. The algorithm divides the population into producers, which are responsible for exploring new food sources, and scroungers, which follow the producers to obtain food. A fraction of sparrows are also assigned as scouts that perform anti-predation behaviors. SSA has attracted attention in feature selection due to its fast convergence, simple structure, and few control parameters.

Several binary variants of SSA have been developed for feature selection. Gad et al. [11] proposed an improved binary SSA (iBSSA) for feature selection in data classification. Liang et al. [24] introduced an enhanced sparrow search swarm optimizer (ESSSO) that integrates multiple search strategies for high-dimensional continuous optimization problems. Tang et al. [37] proposed a chaotic SSA (CSSA) that introduces chaotic maps to enhance population diversity and escape local optima. Despite these improvements, binary SSA variants for feature selection share the same fundamental limitation: they rely on transfer functions to convert continuous positions into binary decisions.

2.3. Limitations of Existing Approaches

The most prevalent approach for binarizing continuous swarm intelligence algorithms is the use of transfer functions. Sigmoid functions [19], V-shaped functions [26], and U-shaped functions [28] have been proposed to map continuous values to selection probabilities. However, these functions impose an inherent constraint: when a continuous position is near zero, the selection probability approaches 0.5, meaning each feature has roughly equal chances of being selected or excluded. In high-dimensional feature selection, where the goal is to identify a small discriminative subset from thousands of candidates, this 0.5 bottleneck severely limits the algorithm's ability to eliminate irrelevant features effectively.

To address this issue, several alternative strategies have been explored. Sun et al. [36] developed a binary SSA (BSSFS) for feature selection, yet it still relies on transfer functions for binarization and thus inherits the probability bottleneck. Huang et al. [18] introduced bIRIME to mitigate the exploitation–exploration imbalance of the RIME optimizer for feature selection, but its binarization is likewise transfer-function based. Shikoun et al. [32] proposed a binary crayfish optimization algorithm (BinCOA) for feature selection, and Beheshti [3] combined time-varying sine and V-shaped transfer functions in a binary marine predators algorithm; both still convert continuous positions through fixed mappings and cannot decisively eliminate irrelevant features.

Another limitation of existing SSA variants is the fixed producer–scrounger ratio. In standard SSA, approximately 20% of the population serves as producers throughout the entire evolutionary process. This static allocation cannot adapt to different search stages: early iterations may benefit from more producers for broader exploration, while later iterations may require more scroungers for fine-grained exploitation. Several adaptive strategies have been proposed for related algorithms. Meng et al. [25] proposed the bird swarm algorithm, which employs adaptive control of flight behaviors to balance exploration and exploitation. Mirjalili et al. [27] proposed the grey wolf optimizer, whose exploration–exploitation balance is controlled by a parameter that decays linearly over iterations. Wang et al. [39] combined multiple search strategies in a multi-strategy SSA variant, demonstrating that hybrid approaches can improve performance on complex optimization problems.

These observations motivate the design of the proposed EDP-SSA, which addresses all three limitations simultaneously through a dynamic feature probability mechanism, an elite consensus tuning strategy, and a time-varying adaptive producer ratio.

3. Proposed EDP-SSA Algorithm

3.1. Overview

The proposed EDP-SSA restructures the binary mapping pipeline of conventional SSA-based feature selection methods. Instead of converting continuous positions to binary decisions through transfer functions, EDP-SSA maintains a dynamic feature probability (DFP) vector $\mathbf{P} = [P_1, P_2, \dots, P_D]$ of dimension D, where each P_j represents the selection probability of the j-th feature. Binary masks are generated directly by Bernoulli sampling: $X_{i,j} \sim \text{Bernoulli}(P_j)$ for the i-th individual and j-th feature. This design allows the probability vector to evolve independently of the continuous positions used by the SSA update equations, enabling more precise control over feature selection behavior.

The overall framework of EDP-SSA is illustrated in Fig. 1. The algorithm consists of four key components: (1) a DFP mechanism with frequency-based feedback that adjusts selection probabilities according to observed feature usage; (2) an elite consensus tuning (ECT) strategy with delayed activation that reinforces features consistently selected by historical best solutions; (3) a time-varying adaptive producer ratio that linearly increases from 15% to 35% during evolution; and (4) a greedy local search post-processing step that refines the final feature subset through single-feature flipping.

3.2. Fitness Evaluation

Each candidate solution is represented as a binary mask $\mathbf{x} = [x_1, x_2, \dots, x_D] \in \{0,1\}^D$, where $x_j = 1$ indicates that the j-th feature is selected and $x_j = 0$ indicates exclusion. The quality of each mask is evaluated using a fitness function that simultaneously considers classification accuracy and feature subset size:

$$\text{Fitness}(\mathbf{x}) = \alpha \cdot E(\mathbf{x}) + (1-\alpha) \cdot \frac{|\mathbf{x}|}{D}, \tag{1}$$

where $E(\mathbf{x})$ denotes the classification error rate computed via K-fold cross-validation with a k-nearest neighbors (KNN) classifier ($k = 5$), $|\mathbf{x}| = \sum_{j=1}^{D} x_j$ is the number of selected features, D is the total number of features, and $\alpha \in [0,1]$ is a user-defined weight that controls the trade-off between accuracy and feature reduction. In this study, $\alpha = 0.8$ is adopted following common practice in the literature [23, 41], giving higher priority to classification accuracy while still encouraging compact feature subsets. A lower fitness value indicates a better solution.

3.3. Probability Bottleneck of Transfer Functions

Conventional binary SSA variants convert the continuous position of each feature into a selection probability through a transfer function and then apply probabilistic rounding [26]. For the widely used S-shaped family,

$$T(x) = \frac{1}{1+e^{-x}}, \tag{2}$$

the mapping satisfies the anti-symmetry property $T(-x) = 1 - T(x)$ with $T(0) = 0.5$. If the continuous positions follow a distribution symmetric around zero, then

$$\begin{aligned} \mathbb{E}[T(x)] &= \int_{-\infty}^{\infty} T(x)\, f(x)\, dx \\ &= \frac{1}{2}\int_{-\infty}^{\infty} [T(x) + T(-x)]\, f(x)\, dx = \frac{1}{2}, \end{aligned} \tag{3}$$

i.e., every feature is expected to be selected with probability 0.5 regardless of its relevance. Two properties of the SSA update equations keep this symmetry intact in high-dimensional spaces. First, positions are initialized uniformly in a symmetric interval (in our implementation, $[-6, 6]$), and the producer/scrounger perturbations are zero-mean and symmetric (the Gaussian walk Q and the random vector $A \in \{-1, +1\}^D$ in Eqs. 13–14). Second, the scrounger step $\left|x_{i,j} - x_{p,j}\right| A_j / D$ shrinks with the feature dimension D, so with D in the thousands the coordinate distribution remains nearly symmetric around zero for a long time, and $T(x)$ stays close to 0.5 because the sigmoid saturates only when $|x|$ exceeds approximately 4. Consequently, transfer-function-based binarization generates dense, nearly random masks, and the swarm must fight an inherent bias toward selecting roughly half of the features. This prediction is consistent with the empirical FSR of the transfer-function-based SSA variants in Tables 5 and 6 (e.g., 0.42–0.49 for basic SSA, iBSSA, and ESSSO), and it explains why these methods stagnate at moderate accuracy on high-dimensional microarray data. The DFP mechanism proposed next removes this bottleneck by evolving the probability vector directly, so each coordinate is free to drift toward 0 or 1 based on empirical evidence instead of being pinned near 0.5 by the geometry of the transfer function.

3.4. Dynamic Feature Probability (DFP)

The DFP mechanism is the key component of EDP-SSA, replacing the conventional transfer-function-based binarization with a direct probability evolution scheme. At the beginning of the algorithm, each feature is assigned a low initial selection probability:

$$P_j^{(0)} = P_{\text{init}}, \quad j = 1, 2, \dots, D, \tag{4}$$

where $P_{\text{init}} = 0.1$ is adopted in this study. This low initial value reflects the prior assumption that, in high-dimensional feature selection, only a small fraction of features are truly informative.

At each iteration t, binary masks are generated by independent Bernoulli sampling:

$$X_{i,j}^{(t)} = \begin{cases} 1 & \text{if } r_{i,j} < P_j^{(t)}, \\ 0 & \text{otherwise,} \end{cases} \tag{5}$$

where $r_{i,j} \sim U(0,1)$ is a uniformly distributed random number. Unlike transfer function approaches, where the selection probability is determined by the continuous position through a fixed mapping, the DFP values are directly updated based on population-level feedback.

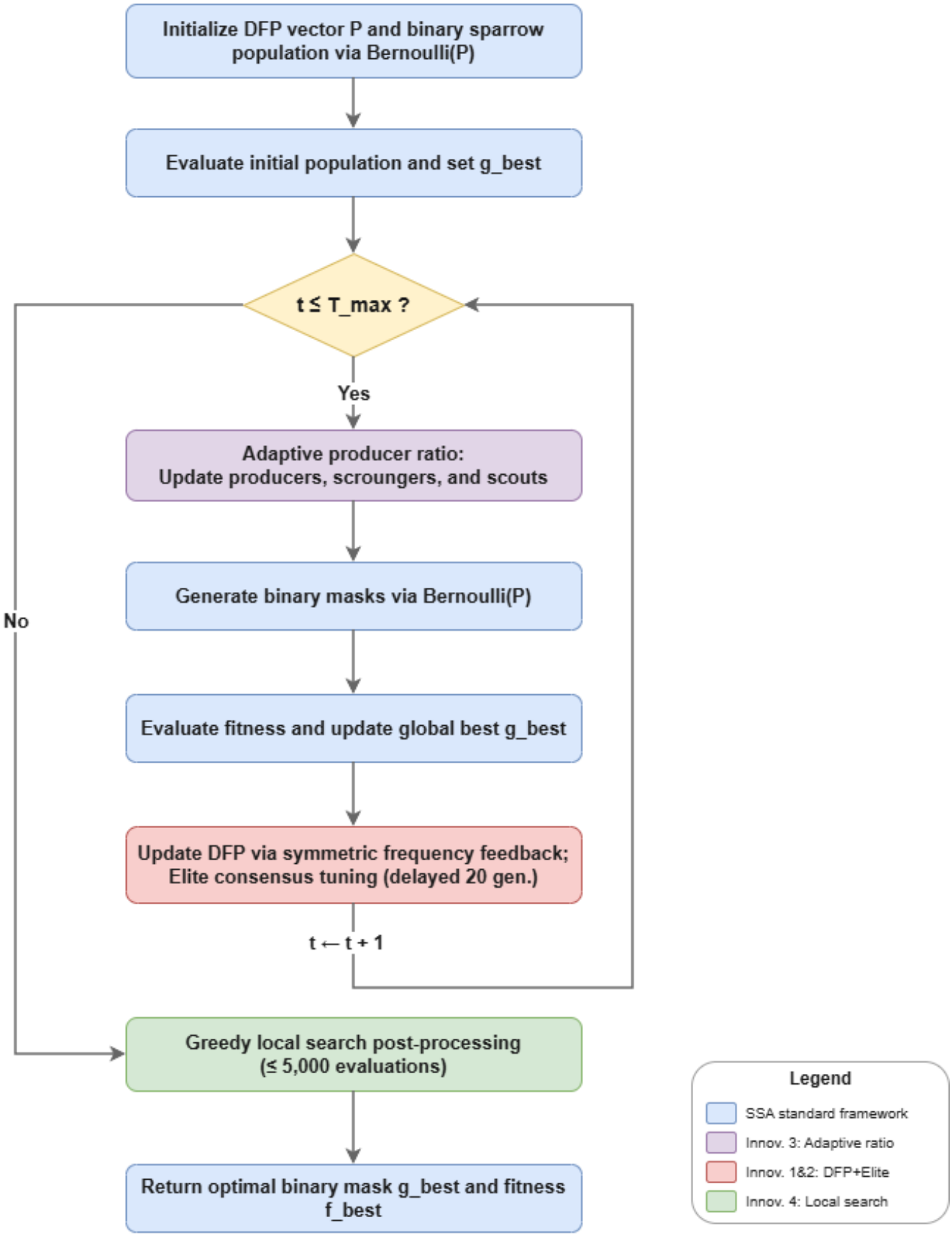


Fig. 1: Flowchart of EDP-SSA.

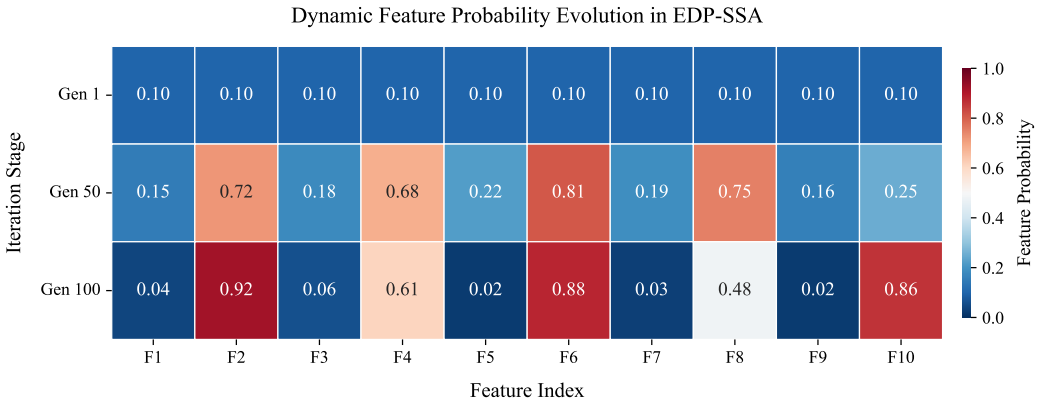


Fig. 2: DFP evolution heatmap.

After evaluating all individuals, the selection frequency of each feature is computed as:

$$f_j^{(t)} = \frac{1}{N} \sum_{i=1}^{N} X_{i,j}^{(t)}, \tag{6}$$

where N is the population size. The DFP vector is then updated according to the following rule:

$$P_j^{(t+1)} = \begin{cases} \max(P_{\min},\ P_j^{(t)} - lr) & \text{if } f_j^{(t)} > f_{\text{high}}, \\ \max(P_{\min},\ P_j^{(t)} - lr) & \text{if } f_j^{(t)} < f_{\text{low}}, \\ P_j^{(t)} & \text{otherwise}, \end{cases} \tag{7}$$

where $f_{\text{high}} = 0.8$ and $f_{\text{low}} = 0.2$ are the high- and low-frequency thresholds, $lr = 0.05$ is the learning rate, and $P_{\min} = 0.01$ and $P_{\max} = 0.95$ are the probability bounds. A single symmetric learning rate is deliberately adopted for both branches: preliminary experiments with asymmetric settings (a larger rate for low-frequency features to accelerate their elimination) did not bring consistent gains across datasets, whereas the single-parameter formulation simplifies the design and makes the sensitivity analysis in Section 4 more tractable.

The rationale behind this update rule is as follows. The DFP mechanism acts exclusively as a pruning operator: it never increases a probability on its own. Features with high selection frequency ($f_j > f_{\text{high}}$) are already included in most candidate solutions. Reducing their probabilities is an exploration-oriented penalty rather than a judgment of irrelevance: when a feature is ubiquitous in the population, the swarm has lost diversity with respect to that feature, and a temporary suppression forces the algorithm to re-examine it instead of converging prematurely to a local optimum. This penalty does not harm genuinely important features: they are protected by the elite consensus tuning (ECT) described next. A feature that keeps appearing in the elite solutions is re-reinforced by ECT with an increment of $\Delta_{\text{inc}} = 0.10$ per iteration, which exceeds the DFP decrement of $lr = 0.05$, so the net drift of such features remains positive. Features with low selection frequency ($f_j < f_{\text{low}}$) are rarely selected and likely irrelevant; further reducing their probabilities accelerates their elimination. Features with moderate frequency remain unchanged, allowing the algorithm to continue evaluating their usefulness. Fig. 2 illustrates the evolution of the DFP vector across iterations on a representative dataset, showing how informative features gradually acquire higher probabilities while irrelevant ones are suppressed.

The convergence behavior of the DFP vector can also be characterized analytically. Without ECT, every coordinate of $\mathbf{P}$ is monotone non-increasing: Eq. 7 either keeps P_j unchanged or reduces it by lr, and P_j is bounded below by $P_{\min}$. By the monotone convergence theorem, each P_j therefore converges to a limit in $[P_{\min}, P_j^{(0)}]$, so the probability vector cannot oscillate. When ECT is active, coordinates may additionally increase by Δ_{inc} or decrease by Δ_{dec}, but remain confined to $[P_{\min}, P_{\max}]$. Once the best mask stabilizes in the late stage of the search, the elite pool becomes stationary, the consensus frequencies $\bar{f}_j$ stop changing, and the composite update reaches a fixed point in which informative features are reinforced toward $P_{\max}$ while irrelevant ones are driven toward $P_{\min}$. This polarization—rather than the 0.5 bottleneck of transfer functions—is exactly the behavior visible in Fig. 2.

3.5. Elite Consensus Tuning

While the DFP mechanism provides effective frequency-based probability adjustment, it operates solely on aggregate population statistics and may lack sufficient guidance toward high-quality regions of the search space. To address this, the elite consensus tuning (ECT) strategy exploits historical best solutions to provide macro-level probability correction.

A sliding elite pool $\mathcal{E}$ maintains the best binary masks from the most recent M generations:

$$\mathcal{E}^{(t)} = \{\mathbf{x}_{\text{best}}^{(t-M+1)}, \ldots, \mathbf{x}_{\text{best}}^{(t)}\}, \tag{8}$$

where $M = 3$ is the elite pool size and $\mathbf{x}_{\text{best}}^{(t)}$ denotes the best binary mask at iteration t. The pool is updated at every iteration by appending the current best mask and removing the oldest entry.

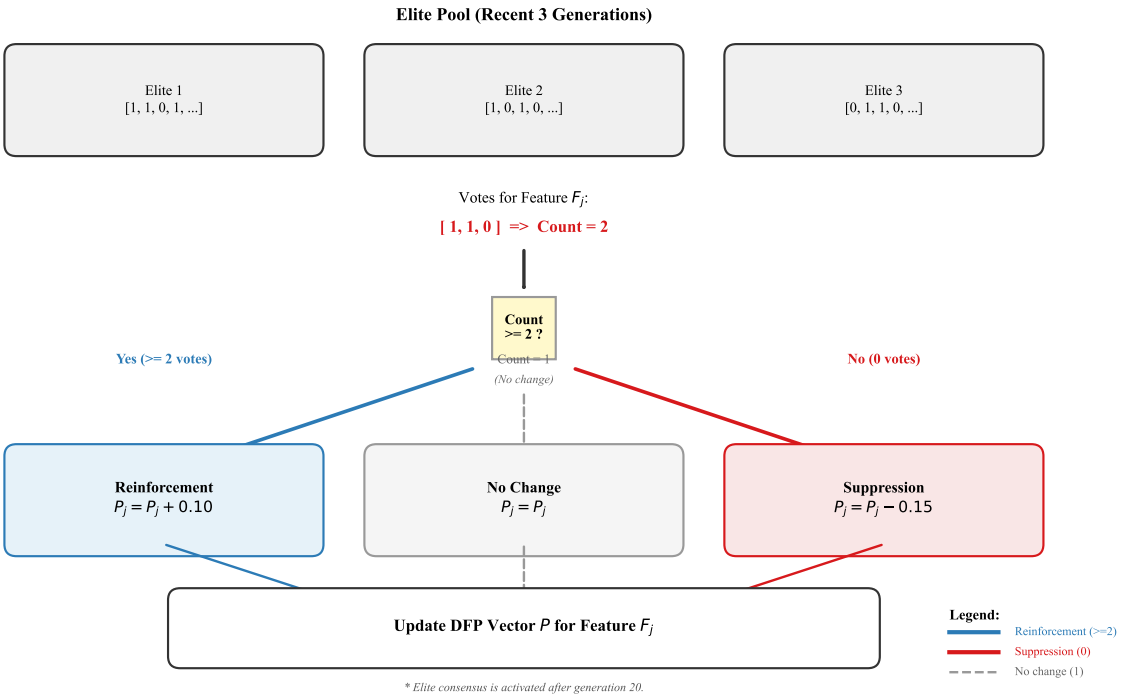


Fig. 3: Elite consensus tuning mechanism.

To avoid premature convergence in the early search stage, the ECT mechanism is activated only after a delay period of $T_{\text{delay}} = 20$ iterations. Once activated, the elite pool frequency is computed as:

$$\bar{f}_j = \frac{1}{M} \sum_{\mathbf{m} \in \mathcal{E}} m_j. \tag{9}$$

Based on $\bar{f}_j$, the DFP vector is adjusted as follows:

$$P_j \leftarrow \begin{cases} \min(P_{\max},\ P_j + \Delta_{\text{inc}}) & \text{if } \bar{f}_j > 0.5, \\ \max(P_{\min},\ P_j - \Delta_{\text{dec}}) & \text{if } \bar{f}_j = 0, \\ P_j & \text{otherwise}, \end{cases} \tag{10}$$

where $\Delta_{\text{inc}} = 0.10$ and $\Delta_{\text{dec}} = 0.15$ are the increment and decrement steps, respectively.

The majority voting rule ($\bar{f}_j > 0.5$) reinforces features that are selected by more than half of the elite solutions, reflecting a consensus among high-quality individuals. The unanimous exclusion rule ($\bar{f}_j = 0$) suppresses features that are never selected by any elite solution, providing a strong signal of irrelevance. The asymmetric step sizes ($\Delta_{\text{dec}} > \Delta_{\text{inc}}$) ensure that irrelevant features are eliminated more aggressively than relevant features are reinforced, maintaining selective pressure toward compact subsets. Fig. 3 provides a visual illustration of this mechanism.

Interaction between DFP and ECT. Within each iteration, the two mechanisms are applied in a fixed order: the frequency-based DFP update (Eq. 7) is executed first, and the consensus-based ECT correction (Eq. 10) is applied afterwards. This ordering makes ECT the final authority: whenever the two mechanisms disagree on a feature, the elite-consensus verdict prevails, which resolves the conflict deterministically rather than through averaging or additional arbitration parameters. The two operators act on largely disjoint regions of the frequency spectrum: DFP penalizes features whose population frequency is extreme (above 0.8 or below 0.2), whereas ECT reinforces features with elite consensus above 0.5 and suppresses features unanimously excluded from the elite pool. Because the DFP decrement ($lr = 0.05$) is smaller than the ECT increment ($\Delta_{\text{inc}} = 0.10$), a feature that is simultaneously over-represented in the population and supported by the elite consensus experiences a net positive drift, which implements the protection mechanism for important features discussed above. The fixed execution order and the magnitude relation $lr < \Delta_{\text{inc}}$ together prevent oscillatory behavior of the probability vector.

3.6. Adaptive Producer Ratio

In standard SSA, the proportion of producers (also called discoverers) is fixed at approximately 20% of the population throughout the entire evolutionary process. This static allocation cannot adapt to different search stages: early iterations benefit from a larger proportion of producers for broader exploration, while later iterations require more scroungers for fine-grained exploitation around promising solutions.

To address this limitation, EDP-SSA employs a time-varying adaptive producer ratio that increases linearly over the course of evolution:

$$R_p(t) = R_p^{\min} + (R_p^{\max} - R_p^{\min}) \cdot \frac{t}{T_{\max}}, \tag{11}$$

where $R_p^{\min} = 0.15$ and $R_p^{\max} = 0.35$ are the lower and upper bounds of the producer ratio, t is the current iteration, and $T_{\max}$ is the maximum number of iterations. The number of producers at iteration t is then:

$$N_p(t) = \max\big(1,\ \min(N-1,\ \mathrm{round}(N \cdot R_p(t)))\big), \quad (12)$$

where N is the population size.

This design ensures that in the early stage (e.g., $t = 0$), only 15% of the population serves as producers, allowing the majority to act as scroungers that exploit the most promising regions identified so far. As the evolution proceeds, the producer ratio gradually increases to 35%, introducing more exploratory behavior to escape local optima and discover new promising regions. This smooth transition avoids the abrupt changes that discrete switching strategies might cause and provides a natural balance between exploration and exploitation throughout the search process.

The standard SSA producer and scrounger update equations are retained:

$$x_{i,j}^{(t+1)} = \begin{cases} x_{i,j}^{(t)} \cdot \exp\left(-\frac{i+1}{\alpha_s \cdot (T_{\max}+\epsilon)}\right) & \text{if } R_2 < ST, \\ x_{i,j}^{(t)} + Q & \text{otherwise,} \end{cases} \quad (13)$$

where $i \in \{1, \ldots, N_p(t)\}$, $R_2 \in [0,1]$ is a random number, $ST = 0.8$ is the safety threshold, $\alpha_s \in [0,1]$ is a random parameter, $Q \sim \mathcal{N}(0,1)$ is a random walk, and ϵ is a small constant to prevent division by zero.

$$x_{i,j}^{(t+1)} = \begin{cases} Q \cdot \exp\left(\frac{x_{\text{worst}}^{(t)} - x_{i,j}^{(t)}}{i^2+\epsilon}\right) & \text{if } i > N/2, \\ x_p^{(t)} + |x_{i,j}^{(t)} - x_p^{(t)}| \cdot \frac{A}{D} & \text{otherwise,} \end{cases} \quad (14)$$

where $x_p^{(t)}$ is the current best position, $x_{\text{worst}}^{(t)}$ is the current worst position, and A is a vector of random values in $\{-1, +1\}$.

3.7. Local Search Post-processing

After the evolutionary process terminates, a greedy local search procedure is applied to the best binary mask to further refine the selected feature subset. The local search operates by iteratively flipping a single feature (changing x_j from 0 to 1 or from 1 to 0) and evaluating the resulting fitness.

At each step, a random permutation of all D features is generated, and each feature is flipped one at a time. If flipping feature j improves the fitness, the change is accepted immediately and the procedure restarts from a new random permutation. If no improvement is found after a complete pass through all features, the local search terminates. To prevent excessive computation on high-dimensional datasets, the total number of fitness evaluations during local search is bounded by $E_{\max} = 5000$. The procedure is summarized as follows:

1. Initialize $\mathbf{x}^* \leftarrow \mathbf{x}_{\text{best}}$, $f^* \leftarrow \text{Fitness}(\mathbf{x}^*)$, count $\leftarrow 1$.
2. Generate a random permutation π of $\{1, 2, \ldots, D\}$.
3. For each $j \in \pi$: flip x_j^*, evaluate fitness, if improved accept and go to Step 2; otherwise revert.
4. If count $< E_{\max}$, go to Step 2; otherwise return $\mathbf{x}^*$.

This single-feature flipping neighborhood is computationally efficient and has been shown to be effective for binary feature selection problems [22].

It is important to clarify the evaluation protocol behind the local search to rule out overfitting concerns. The local search optimizes exactly the same fitness function as the evolutionary search phase, i.e., 5-fold stratified cross-validation with a fixed fold partition (a fixed random state), so it never observes the final evaluation folds. The final reported accuracy of every algorithm—including EDP-SSA after local search—is always recomputed with an independent 10-fold stratified cross-validation. The bounded evaluation budget ($E_{\max} = 5000$) further limits the degree to which the local search can adapt to the search-phase partitions. In the ablation study, variant (iii) (SSA+DFP+Elite) excludes both the adaptive producer ratio and the local search, so its gap of 2.69 percentage points to the full EDP-SSA quantifies the combined contribution of these two components. To isolate the contribution of the local search alone, we additionally ran the full model with the local-search post-processing disabled on three representative datasets (colon, SRBCT, and lung) under the same 10 random seeds. The local search adds an average of 1.45 percentage points of accuracy on these datasets (from 94.01% to 95.46%), with the largest gain of 2.82 percentage points on SRBCT and the smallest of 0.39 percentage points on lung, which confirms that the fourth component contributes measurably to the final performance without inflating the reported accuracy beyond its legitimate improvement.

3.8. Complete Algorithm

The complete EDP-SSA algorithm is summarized in Algorithm 1. The algorithm begins by initializing the DFP vector at a low value (0.1) for all features and generating the initial population of sparrow continuous positions randomly (Lines 1–3). In each iteration, the adaptive producer ratio is computed and the producer/scrounger positions are updated (Line 5), after which binary masks are generated via Bernoulli sampling from the DFP vector and evaluated using the fitness function (Line 6). The DFP vector is then updated via frequency-based feedback (Line 7), and the current best mask is appended to the elite pool (Line 8); if the delay period has passed, the elite consensus tuning adjusts the DFP vector based on majority voting (Lines 9–10). Note that ECT is applied after the DFP update within each iteration, so the elite consensus acts as the final correction whenever the two mechanisms disagree. After the maximum number of iterations is reached, a greedy local search is applied to refine the best mask (Line 12), and the algorithm returns the best mask and the converged DFP vector (Line 13).

3.9. Computational Complexity Analysis

The computational cost of EDP-SSA per iteration is dominated by fitness evaluation, which is inherent to wrapper-based feature selection. Let N, D, and $T_{\max}$ denote the population size, the feature dimension, and the maximum

Algorithm 1 EDP-SSA

Require: $X, y, N, T_{\max} = 100, T_{\text{delay}} = 20$
Initialize $\mathbf{P} = 0.1$ for all D dimensions
Initialize sparrow continuous positions randomly
$\mathcal{E} \leftarrow \emptyset$
for $t = 1$ to $T_{\max}$ **do**
    Compute $R_p(t)$; update producer/scrounger positions (Eqs. 13–14)
    Generate masks $\mathbf{X}$ via $\mathbf{X}_{i,j} \sim \text{Bernoulli}(P_j)$ and evaluate fitness
    Update $\mathbf{P}$ via DFP frequency feedback (Eq. 7)
    Append best mask to elite pool $\mathcal{E}$
    if $t > T_{\text{delay}}$ **then**
        Update $\mathbf{P}$ via elite consensus tuning (Eq. 10)
    end if
end for
Apply greedy local search to best mask
return best mask and $\mathbf{P}$

number of iterations, respectively, and let T_{fit} denote the cost of a single fitness evaluation (one 5-fold KNN cross-validation). In each iteration, the following operations are performed: (i) Bernoulli mask generation, which costs $O(ND)$; (ii) fitness evaluation of all individuals, which costs $O(N \cdot T_{\text{fit}})$; (iii) elite pool maintenance and elite consensus tuning, which cost $O(MD + D)$ with $M = 3$; (iv) producer/scrounger position updates, which cost $O(ND)$; and (v) DFP frequency statistics and probability updates, which cost $O(ND+D)$. The overall per-iteration complexity is therefore $O(ND + N \cdot T_{\text{fit}})$, and the total evolutionary cost is $O\big(T_{\max} \cdot (ND + N \cdot T_{\text{fit}})\big)$. The greedy local search adds at most $E_{\max} = 5000$ extra fitness evaluations, i.e., $O(E_{\max} \cdot T_{\text{fit}})$. Compared with the basic SSA, whose per-iteration complexity is also $O(ND + N \cdot T_{\text{fit}})$, the DFP and ECT mechanisms introduce only low-order $O(ND+D)$ overhead; the asymptotic complexity is therefore unchanged, which is consistent with the comparable runtimes reported in Tables 9 and 10.

4. Experiments and Discussion

4.1. Experimental Setup

4.2. Datasets and Comparison Algorithms

Datasets. The proposed algorithm is evaluated on 16 widely used high-dimensional microarray datasets, summarized in Table 1. These datasets cover tumor classification (11_Tumors, SRBCT), leukemia subtyping (ALLAML, Leukemia-3, leukemia, CLL_SUB_111), lung cancer (lung, SMK_CAN_187), and other cancer types (colon, GLI_85, Prostate_GE, DLBCL, TOX_171, and arcene). The number of samples ranges from 62 (colon) to 468 (AP_Breast_Uterus and Leukemia-3), while the number of original features varies from 2,000 (colon) to 22,283 (GLI_85). The number of classes ranges from 2 (binary classification) to 11 (multi-class tumor classification), providing a diverse benchmark for evaluating feature selection algorithms. All datasets are preprocessed with standard normalization to zero mean and unit variance.

Table 1
Summary of the 16 high-dimensional datasets.

No.	Dataset	Samples	Features	Classes	Ref.
1	colon	62	2000	2	[1]
2	leukemia	72	7070	2	[12]
3	lung	203	3312	5	[13]
4	GLI_85	85	22283	2	[29]
5	ALLAML	72	7129	2	[2]
6	Prostate_GE	102	5966	2	[34]
7	DLBCL	77	5469	2	[33]
8	SRBCT	83	2308	4	[20]
9	11_Tumors	174	12533	11	[31]
10	AP_Breast_Uterus	468	10936	2	[9]
11	Leukemia-3	468	10936	2	[9]
12	AP_Lung_Kidney	386	10936	2	[9]
13	SMK_CAN_187	187	19993	2	[35]
14	CLL_SUB_111	111	11340	3	[9]
15	arcene	200	10000	2	[15]
16	TOX_171	171	5748	4	[9]

Table 2
Default parameters of EDP-SSA.

Parameter	Value
Population size N	100
Maximum iterations $T_{\max}$	100
Fitness weight α	0.8
k for KNN	5
Search CV folds	5
Final evaluation CV folds	10
Seeds	$\{7, 11, 13, 17, 19, 23, 29, 31, 37, 41\}$
lr (DFP step)	0.05
Elite pool size	3
Elite delay T_{delay}	20
Producer ratio R_p range	$[0.15, 0.35]$
Local search max evaluations	5000

Comparison algorithms. EDP-SSA is compared against 10 feature selection algorithms, which can be divided into two groups. The first group comprises three SSA-based methods: the basic SSA [42], improved binary SSA (iBSSA) [11], and enhanced sparrow search swarm optimizer (ESSSO) [24]. The second group includes seven other well-established methods: binary particle swarm optimization (BPSO) [19], binary grey wolf optimizer (BGWO) [8], binary PSO with V-shaped transfer function (BPSO-V) [26], binary crayfish optimization algorithm (BinCOA) [32], improved binary RIME algorithm (bIRIME) [18], binary marine predators algorithm with time-varying sine and V-shaped transfer functions (BMPA-TVSinV) [3], and support vector machine recursive feature elimination (SVM-RFE) [16].

Parameter settings. For fair comparison, all metaheuristic algorithms share the following common parameters: population size $N = 100$, maximum iterations $T_{\max} = 100$, and fitness weight $\alpha = 0.8$. The algorithm-specific parameters of the competing methods (e.g., the inertia weight and acceleration coefficients of BPSO, the transfer function settings of BGWO and BPSO-V) follow

the values recommended in their original papers, and no dataset-specific tuning is performed for any algorithm. The parameters of EDP-SSA are fixed once for all datasets, based on the sensitivity analysis reported in Section 4. The KNN classifier with $k = 5$ is used for fitness evaluation during the search phase with 5-fold stratified cross-validation [5]. Final performance assessment employs 10-fold stratified cross-validation to provide a more reliable accuracy estimate. Each algorithm is executed with 10 different random seeds $\{7, 11, 13, 17, 19, 23, 29, 31, 37, 41\}$, and the mean and standard deviation are reported. The default parameters specific to EDP-SSA are listed in Table 2.

Implementation environment. All algorithms are implemented in Python 3.10 with the scikit-learn library for the KNN classifier and cross-validation procedures, and the SciPy library for the statistical tests. The experiments are conducted on a server equipped with a 32-core CPU and 64 GB of RAM.

Evaluation metrics. Three main metrics are used to evaluate the performance: (1) classification accuracy, computed as the proportion of correctly classified samples; (2) feature selection ratio (FSR), defined as the ratio of selected features to the total number of features; and (3) fitness value, computed according to Eq. 1. Additionally, runtime is recorded to assess computational efficiency.

4.3. Overall Performance Comparison

Classification accuracy. Tables 3 and 4 present the mean and standard deviation of classification accuracy for all algorithms across the 16 datasets. EDP-SSA achieves the best average accuracy of 0.9188, outperforming all competing methods. Among the SSA-based methods (Table 3), EDP-SSA surpasses the basic SSA (0.8530), iBSSA (0.8610), and ESSSO (0.8609) by margins of 6.58, 5.78, and 5.79 percentage points, respectively. Among the other comparison methods (Table 4), the second-best average accuracy is achieved by bIRIME (0.8927), followed by BPSO (0.8657), BinCOA (0.8629), BMPA-TVSinV (0.8628), and BGWO (0.8624). EDP-SSA outperforms bIRIME by 2.61 percentage points on average. EDP-SSA shows its largest gains on the most challenging benchmarks: on CLL_SUB_111, it achieves 80.07% accuracy, exceeding the second-best method bIRIME (69.64%) by 10.43 percentage points—the widest margin observed across all datasets. On SRBCT, a four-class problem, EDP-SSA reaches 97.64%, compared to 95.68% for bIRIME. On the colon dataset, EDP-SSA attains 93.36%, outperforming bIRIME (89.76%) by 3.60 percentage points. On the SMK_CAN_187 dataset, EDP-SSA achieves 75.40%, surpassing all competing methods by at least 3.3 percentage points. The only dataset where EDP-SSA does not achieve the best result is lung, where bIRIME (97.59%) and BinCOA (96.89%) slightly outperform EDP-SSA (96.80%); the difference is marginal (0.79 and 0.09 percentage points, respectively). The "+" markers in Tables 3 and 4 indicate that EDP-SSA is significantly better than the corresponding algorithm according to the Wilcoxon signed-rank test at the 0.05 significance level [6, 40].

Feature selection ratio. Tables 5 and 6 report the feature selection ratio (FSR) results. EDP-SSA achieves the lowest average FSR of 0.0104 (i.e., selecting only 1.04% of features on average), which is substantially lower than all competing methods. Among the SSA-based methods (Table 5), the basic SSA selects 42.18% of features, iBSSA selects 37.49%, and ESSSO selects 49.33%. Among the other methods (Table 6), SVM-RFE achieves the second-lowest FSR of 1.67%, followed by bIRIME (7.98%) and BMPA-TVSinV (9.85%). However, SVM-RFE achieves this low FSR at the cost of significantly lower accuracy (average 81.88%), whereas EDP-SSA simultaneously achieves the highest accuracy with the lowest FSR.

On individual datasets, EDP-SSA compresses the feature space substantially. On the colon dataset, EDP-SSA selects only 18 features from 2,000 original features (FSR = 0.009) while achieving 93.36% accuracy. On ALLAML, it selects 60 features from 7,129 (FSR = 0.0084) with 96.29% accuracy. On DLBCL, only 40 features are selected from 5,469 (FSR = 0.0073) with 96.13% accuracy. On SRBCT, EDP-SSA selects merely 24 features from 2,308 (FSR = 0.0104) while achieving the highest accuracy of 97.64%. These results confirm that the DFP mechanism effectively drives selection probabilities toward zero for irrelevant features, enabling the algorithm to identify extremely compact yet highly discriminative feature subsets.

Fitness value. Tables 7 and 8 compare the fitness values, which jointly consider accuracy and feature subset size. EDP-SSA achieves the best average fitness of 0.0436, substantially lower than all competing methods. The nearest competitor is bIRIME with an average fitness of 0.0794; however, this comparison must be interpreted carefully because bIRIME selects far more features (FSR = 7.98%), which inflates the feature reduction term in the fitness function. Among the SSA-based methods, EDP-SSA (0.0436) dramatically outperforms SSA (0.1926), iBSSA (0.1769), and ESSSO (0.2007), confirming that the DFP mechanism and elite consensus tuning significantly improve the multi-objective optimization capability.

Runtime analysis. Tables 9 and 10 compare the runtime (in seconds) of all algorithms. EDP-SSA requires an average runtime of approximately 482 seconds, which is comparable to other metaheuristic wrapper methods. SSA-based methods range from 195 seconds (ESSSO) to 669 seconds (basic SSA), while other methods range from a few seconds (SVM-RFE) to 841 seconds (bIRIME on GLI_85). SVM-RFE is the fastest method (average 8.93 seconds) due to its deterministic ranking-and-elimination nature, but at the cost of substantially lower accuracy. The runtime of EDP-SSA is primarily determined by the fitness evaluations (KNN cross-validation) at each iteration, and the local search post-processing adds a bounded overhead of at most 5,000 evaluations. Given that EDP-SSA achieves the best accuracy and FSR simultaneously, the computational cost is justified.

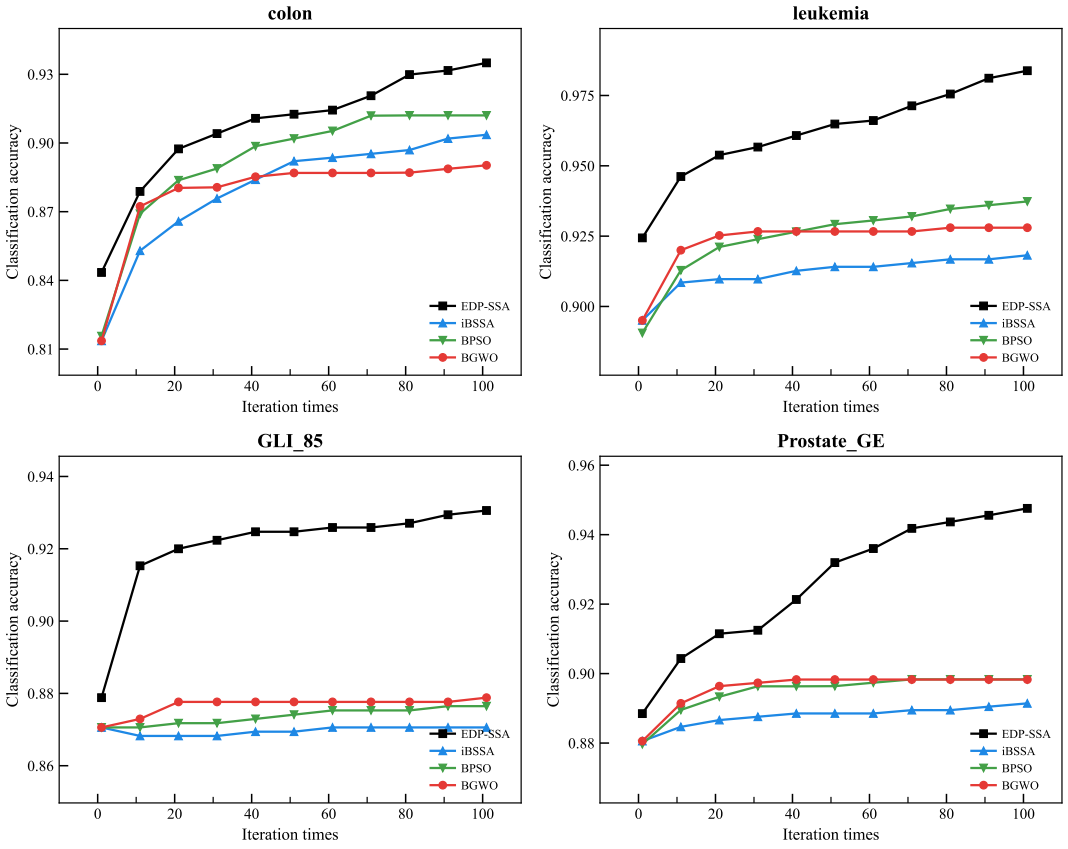


Fig. 4: Comparison of the accuracy of each comparison algorithm.

4.4. Statistical Significance Analysis

To rigorously assess whether the performance improvements of EDP-SSA over competing algorithms are statistically significant, the two-sided Wilcoxon signed-rank test [40] is applied at the 0.05 significance level. For each dataset, the 10 accuracy values of EDP-SSA (one per independent run) and the 10 accuracy values of each competing algorithm form two paired samples of size 10, and the test is performed on these paired vectors rather than on the aggregate means. The Wilcoxon signed-rank test is a non-parametric statistical test suitable for comparing two related samples, and it does not require the assumption of normal distribution. The null hypothesis is that there is no significant difference between the two algorithms.

In Tables 3 and 4, the "+" marker indicates that EDP-SSA is significantly better than the corresponding algorithm at the $p < 0.05$ level, the "≈" marker indicates no statistically significant difference, and the "−" marker indicates that EDP-SSA is significantly worse. Across all 16 datasets and 10 competing algorithms, a total of 160 pairwise comparisons are conducted, among which EDP-SSA achieves statistically significant wins in 140 cases, ties in 19 cases, and a significant loss in only 1 case. Among the SSA-based comparisons (Table 3), EDP-SSA is significantly better on all 16 datasets against SSA Basic, and on 13 out of 16 datasets against both iBSSA and ESSSO, with the remaining comparisons being ties; no significant loss occurs within this group. Among the other comparisons (Table 4), EDP-SSA significantly outperforms bIRIME on 11 out of 16 datasets and ties on 4 datasets (DLBCL, SRBCT, arcene, and leukemia); the only dataset on which EDP-SSA is significantly worse is lung, where bIRIME achieves 97.59% versus 96.80% for EDP-SSA. Against BPSO, BGWO, BPSO-V, BinCOA, BMPA-TVSinV, and SVM-RFE, EDP-SSA achieves significant wins or ties on every dataset, with no significant losses.

The Friedman test is also conducted as a non-parametric multiple comparison test across all 16 datasets simultaneously, treating the 11 algorithms as 11 treatments [10]. The average rank of EDP-SSA is 1.125 (the best among all 11 algorithms), followed by bIRIME (2.000), BPSO (4.875), BMPA-TVSinV (5.406), and BinCOA (5.469). The basic SSA ranks 9.219 and SVM-RFE ranks last at 9.438, confirming that the proposed improvements substantially elevate the algorithm's performance. The Friedman statistic is $\chi^2 = 105.952$ with $df = 10$, yielding $p < 0.001$ and rejecting the null hypothesis that all algorithms perform equally.

Since the Friedman test rejects the null hypothesis, the Nemenyi post-hoc test [6] is further applied to identify which pairs of algorithms differ significantly. With $k = 11$ algorithms and $N = 16$ datasets, the critical difference at $\alpha = 0.05$ is $CD = q_{0.05}\sqrt{k(k+1)/(6N)} = 3.219 \times$

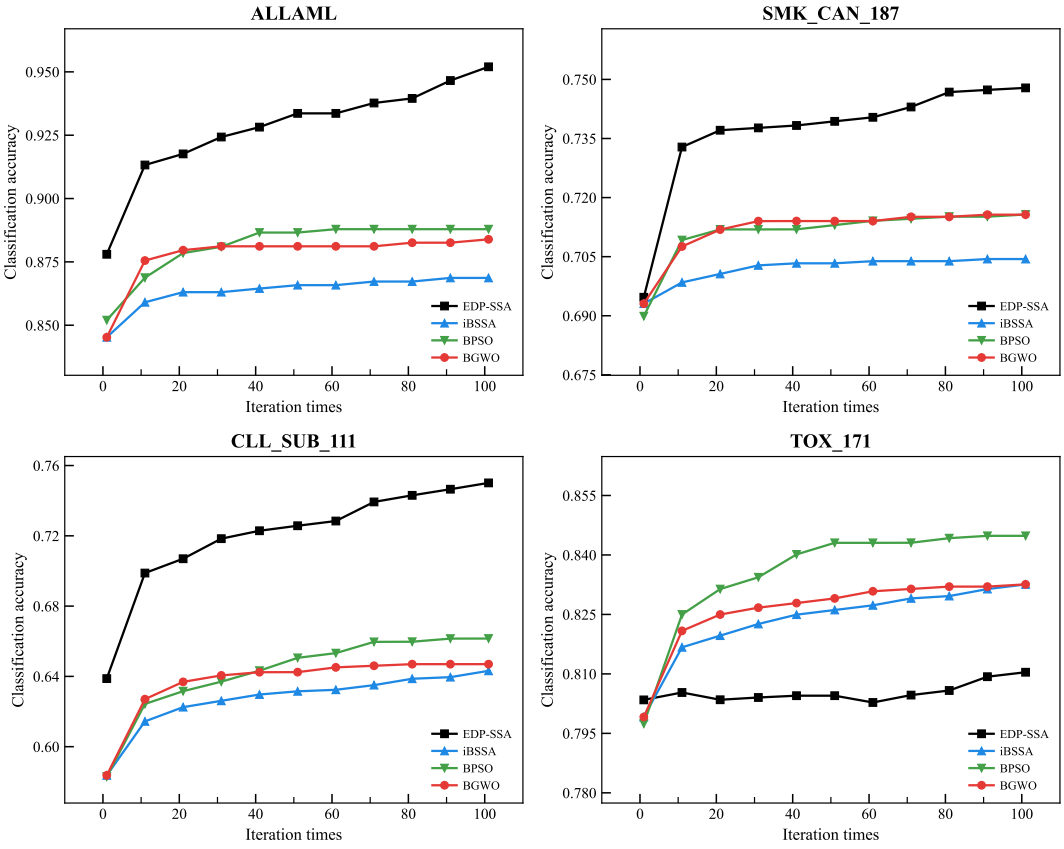


Fig. 5: Comparison of the accuracy of each comparison algorithm.

$\sqrt{132/96} \approx 3.78$, where $q_{0.05} = 3.219$ is the critical value of the studentized range statistic for $k = 11$. Fig. 13 presents the critical difference diagram. EDP-SSA is significantly better than 8 of the 10 competitors (BMPA-TVSinV, BinCOA, BGWO, ESSSO, iBSSA, BPSO-V, basic SSA, and SVM-RFE), whose rank differences from EDP-SSA all exceed the CD. The rank differences of bIRIME (0.875) and BPSO (3.750) remain within the CD; for bIRIME, the per-dataset Wilcoxon test still reports significant wins on 11 of 16 datasets, which is consistent because the Nemenyi test is a conservative test that aggregates over all datasets simultaneously.

These statistical tests confirm that the advantage of EDP-SSA reflects genuine algorithmic differences rather than random fluctuations.

4.5. Convergence, Stability and Sensitivity Analysis

Convergence analysis. Figs. 4–7 display the convergence curves of classification accuracy for all 11 algorithms across the 16 datasets. Each curve plots the best accuracy achieved up to each iteration, averaged over 10 random seeds. The convergence curves reveal several patterns. First, EDP-SSA consistently reaches higher final accuracy than all competing methods on the majority of datasets, particularly on challenging problems such as CLL_SUB_111 (80.07% vs. 69.64% for bIRIME), SRBCT (97.64% vs. 95.68%), and SMK_CAN_187 (75.40% vs. 71.57%). Second, EDP-SSA exhibits rapid convergence in the early iterations (approximately the first 20–30 iterations), which can be attributed to the DFP mechanism quickly identifying and reinforcing informative features through frequency-based feedback. Third, in the later iterations, EDP-SSA continues to show gradual improvement, reflecting the contribution of the elite consensus tuning and adaptive producer ratio in fine-tuning the search. By contrast, most competing methods plateau earlier and fail to achieve further improvement, indicating premature convergence. Among the SSA-based methods, the basic SSA shows the slowest convergence and lowest final accuracy, while iBSSA and ESSSO show moderate improvement but remain far below EDP-SSA. Among the other methods, bIRIME and BMPA-TVSinV show competitive convergence on some datasets but lack the consistency of EDP-SSA across all 16 datasets.

Stability analysis. Fig. 8 presents box plots of the accuracy distributions across 10 independent runs for each algorithm on representative datasets. The box plots reveal the robustness and consistency of each algorithm. EDP-SSA exhibits narrow interquartile ranges and few outliers on most datasets, indicating stable performance across different random initializations. On datasets such as AP_Breast_Uterus

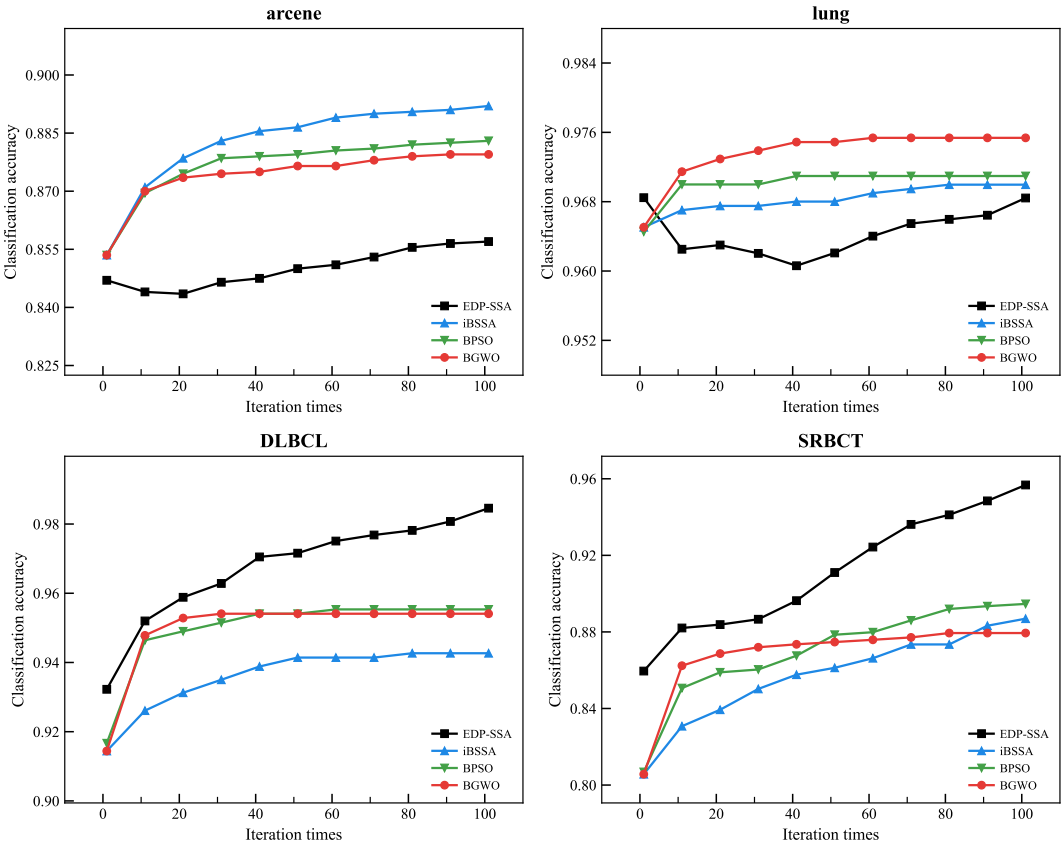


Fig. 6: Comparison of the accuracy of each comparison algorithm.

and AP_Lung_Kidney, EDP-SSA achieves tight distributions with standard deviations of 0.0038 and 0.0039, respectively (see Tables 3 and 4). In contrast, the basic SSA shows wider spreads on the same datasets (std 0.0051 and 0.0046) and notably larger variability on difficult problems such as SRBCT (std 0.0315 vs. 0.0194 for EDP-SSA) and colon (std 0.0222 vs. 0.0225 for EDP-SSA). The stability of EDP-SSA can be attributed to the DFP mechanism, which provides a consistent search direction through probability guidance, and the elite consensus tuning, which reinforces reliable feature selections across generations.

Ablation study. To verify the contribution of each proposed component, an ablation study is conducted by comparing EDP-SSA with several variants: (i) SSA Basic (the original SSA without any proposed mechanism); (ii) SSA+DFP, which uses only the DFP mechanism without elite consensus tuning; (iii) SSA+DFP+Elite, which further incorporates the elite consensus tuning. Fig. 14 visualizes the average accuracy of EDP-SSA and each variant across all 16 datasets.

The ablation results reveal several important findings. First, all DFP-based variants dramatically outperform the basic SSA in both accuracy and feature reduction, confirming that the dynamic feature probability mechanism is the most fundamental contribution of this work. The basic SSA selects an average of 3,943.2 features (an FSR of 42.18%), while all DFP-based variants reduce this to fewer than 106 features on average (an FSR below 1.19%). Second, the full EDP-SSA achieves the highest accuracy among all variants, with an average accuracy of 91.88% across the 16 datasets, compared to 88.03% for the DFP-only variant and 89.19% for the DFP+Elite variant. The DFP mechanism alone raises the average accuracy from 85.30% (basic SSA) to 88.03%, a gain of 2.73 percentage points, while the elite consensus tuning contributes a further 1.16 percentage points, and the adaptive producer ratio with local search together raise the accuracy to 91.88%. Third, the adaptive producer ratio and local search post-processing provide additional gains, particularly on datasets such as colon and leukemia, where the full model selects substantially fewer features (averaging 18.0 and 61.6 features, respectively, compared to 897.2 and 3,286.4 for the basic SSA) while maintaining or improving accuracy. These results confirm that each proposed mechanism is essential to the overall performance of EDP-SSA.

Parameter sensitivity analysis. Fig. 15 presents the parameter sensitivity analysis of EDP-SSA with respect to three key parameters: the DFP learning rate (lr), the elite pool size (M), and the elite delay (T_{delay}). Fig. 15a shows the heatmap of accuracy across different parameter combinations, Fig. 15b shows the corresponding feature count, and Fig. 15c plots accuracy versus feature count. The results indicate that EDP-SSA achieves consistently high accuracy across a wide range of learning rates ($lr \in [0.02, 0.10]$),

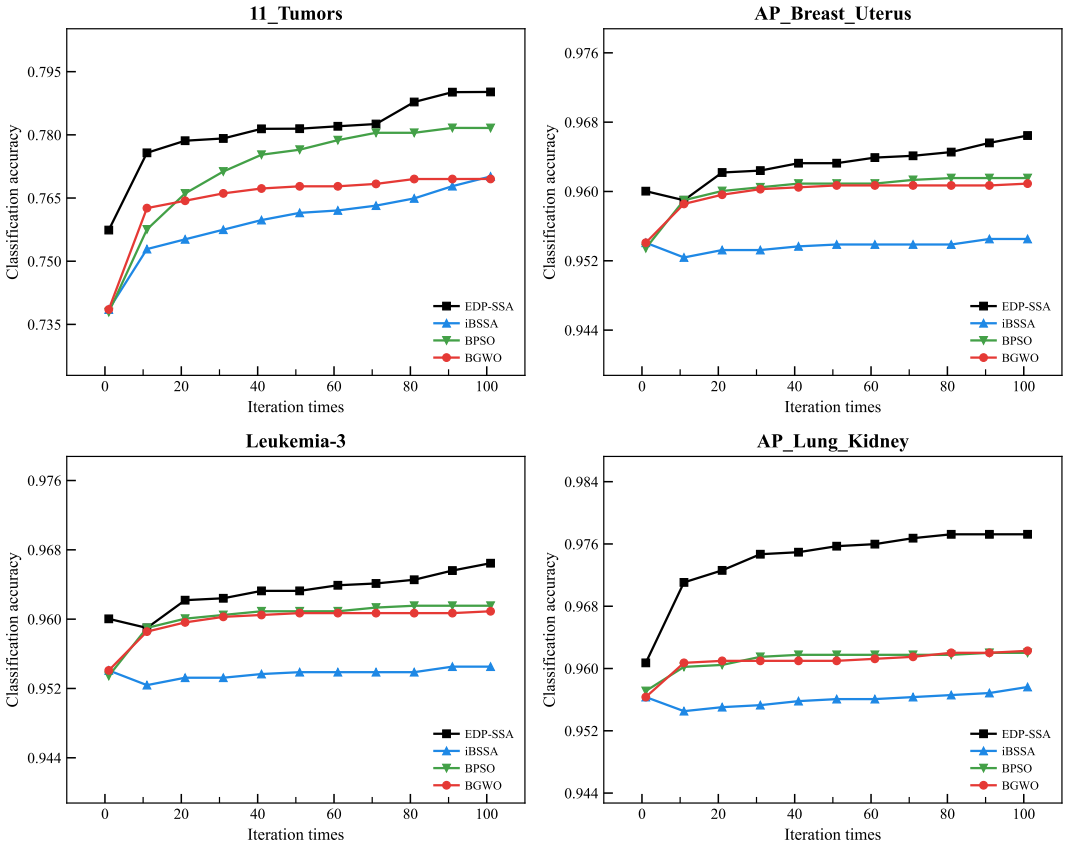


Fig. 7: Comparison of the accuracy of each comparison algorithm.

suggesting that the algorithm is not overly sensitive to this parameter. Very small learning rates ($lr < 0.01$) slow down the probability adaptation and reduce accuracy, while very large rates ($lr > 0.15$) cause excessive oscillation in the probability vector. The elite pool size $M = 2$–4 yields the best results, with $M = 3$ providing the best balance between guidance quality and diversity. The elite delay $T_{\text{delay}} = 15$–25 is recommended, as shorter delays may lead to premature consensus and longer delays may waste valuable iterations without elite guidance. Overall, the parameter sensitivity analysis confirms that EDP-SSA performs robustly with the default parameter settings ($lr = 0.05$, $M = 3$, $T_{\text{delay}} = 20$) adopted in this study. The symmetric setting (the same lr for both frequency branches of Eq. 7) was compared against an asymmetric variant that accelerates the elimination of low-frequency features, and no consistent advantage of the asymmetric variant was observed, supporting the single-parameter design.

5. Conclusion and Future Work

This paper proposed an Elite-Guided Dynamic Feature Probability Sparrow Search Algorithm (EDP-SSA) for high-dimensional feature selection. The algorithm addresses three fundamental limitations of existing binary SSA variants through four complementary mechanisms. First, the dynamic feature probability (DFP) mechanism replaces conventional transfer functions with a directly evolving probability vector, enabling Bernoulli sampling to generate binary masks without the 0.5 probability bottleneck. Second, the elite consensus tuning (ECT) strategy with delayed activation exploits historical best solutions through majority voting, providing robust probability correction that reinforces consistently selected features and suppresses unanimously excluded ones. Third, a time-varying adaptive producer ratio increases linearly from 15% to 35% during evolution, enabling a smooth transition from exploration to exploitation. Fourth, a greedy single-feature flipping local search refines the final feature subset under a bounded evaluation budget. By shifting the binarization paradigm from position-dependent transfer functions to evidence-driven probability evolution, EDP-SSA offers a new perspective for high-dimensional discrete optimization beyond conventional swarm intelligence.

Experiments on the 16 microarray benchmarks against 10 competing algorithms show the effectiveness of the proposed approach. EDP-SSA achieved the highest average classification accuracy of 91.88%, outperforming the second-best method bIRIME (89.27%) by 2.61 percentage points. More importantly, EDP-SSA selected only 1.04% of features on average, compared to 7.98% for the nearest competitor, confirming its superior feature compression capability. The algorithm also obtained the best average

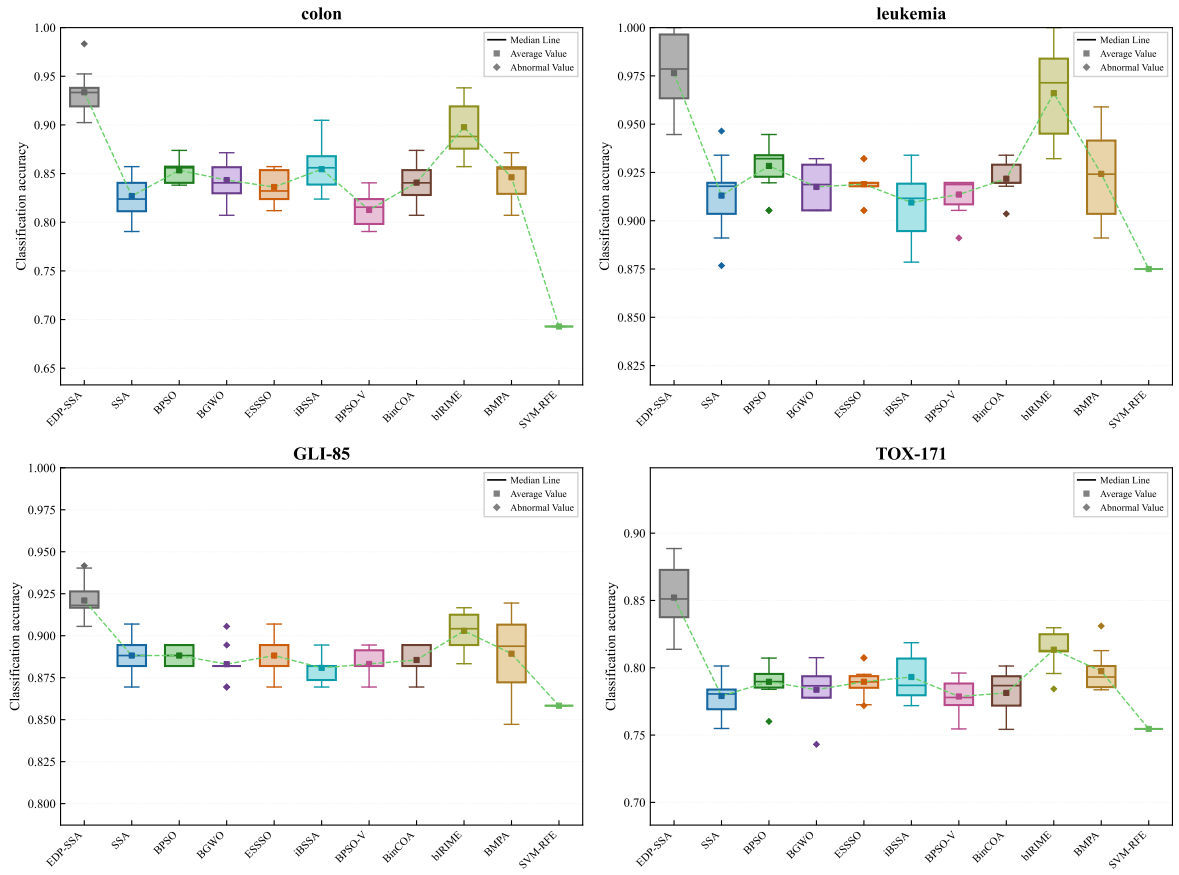


Fig. 8: Stability comparison through box plots.

fitness value of 0.0436 and the best Friedman rank of 1.125 across all 16 datasets. Wilcoxon signed-rank tests confirmed statistically significant improvements in 140 of the 160 pairwise comparisons, with only one significant loss (against bIRIME on lung), and the Nemenyi post-hoc test further confirmed that EDP-SSA is significantly better than 8 of the 10 competitors across all datasets. Ablation experiments on the 16 datasets quantified the progressive contribution of each component: the DFP mechanism raised the average accuracy from 85.30% (basic SSA) to 88.03%, the elite consensus tuning provided an additional improvement to 89.19%, and the full model with the adaptive producer ratio and local search reached 91.88%. Parameter sensitivity analysis demonstrated that EDP-SSA performs robustly across a wide range of parameter settings, with stable accuracy over $lr \in [0.02, 0.10]$.

Despite the promising results, EDP-SSA has certain limitations. The computational cost is higher than filter methods such as SVM-RFE due to the wrapper-based evaluation strategy, which requires repeated KNN cross-validation at each iteration. The current implementation uses a single base classifier (KNN), and the selected feature subsets may not generalize equally well to other classifiers. Additionally, the algorithm is designed for single-objective optimization with a fixed accuracy–FSR trade-off weight ($\alpha = 0.8$).

Two directions deserve further investigation. First, extending EDP-SSA to a multi-objective framework that treats accuracy and feature count as separate objectives, combined with self-adaptive trade-off weights, would provide a Pareto front of solutions and remove the need for manual weight selection. Second, applying the evidence-driven probability paradigm to broader settings—multi-omics data integration and feature selection for deep learning models—offers a natural path beyond the microarray benchmarks considered here.

Table 3
Comparison of EDP-SSA and SSA based methods in terms of average and std of accuracy.

Datasets	Stat. measure	EDP-SSA	SSA based		
			SSA Basic	iBSSA	ESSSO
11_Tumors	mean	**0.8366**	0.7422 (+)	0.7447 (+)	0.7437 (+)
	std	0.0217	0.0104	0.0120	0.0148
ALLAML	mean	**0.9629**	0.8641 (+)	0.8752 (+)	0.8811 (+)
	std	0.0258	0.0194	0.0155	0.0096
AP_Breast Uterus	mean	**0.9752**	0.9562 (+)	0.9532 (+)	0.9570 (+)
	std	0.0038	0.0051	0.0039	0.0038
AP_Lung Kidney	mean	**0.9848**	0.9573 (+)	0.9581 (+)	0.9593 (+)
	std	0.0039	0.0046	0.0021	0.0029
CLL_SUB 111	mean	**0.8007**	0.5977 (+)	0.6134 (+)	0.6245 (+)
	std	0.0307	0.0308	0.0144	0.0123
DLBCL	mean	**0.9613**	0.9280 (+)	0.9404 (≈)	0.9430 (≈)
	std	0.0239	0.0213	0.0112	0.0067
GLI_85	mean	**0.9210**	0.8882 (+)	0.8808 (+)	0.8882 (+)
	std	0.0125	0.0106	0.0090	0.0121
Leukemia-3	mean	**0.9733**	0.9562 (+)	0.9532 (+)	0.9570 (+)
	std	0.0034	0.0051	0.0039	0.0038
Prostate_GE	mean	**0.9423**	0.8668 (+)	0.8726 (+)	0.8709 (+)
	std	0.0147	0.0180	0.0127	0.0047
SMK_CAN 187	mean	**0.7540**	0.6768 (+)	0.6912 (+)	0.6965 (+)
	std	0.0220	0.0187	0.0109	0.0090
SRBCT	mean	**0.9764**	0.8676 (+)	0.9025 (+)	0.8761 (+)
	std	0.0194	0.0315	0.0265	0.0272
TOX_171	mean	**0.8522**	0.7791 (+)	0.7931 (+)	0.7896 (+)
	std	0.0255	0.0139	0.0167	0.0120
arcene	mean	**0.8825**	0.8670 (+)	0.8705 (≈)	0.8670 (≈)
	std	0.0169	0.0101	0.0148	0.0109
colon	mean	**0.9336**	0.8269 (+)	0.8545 (+)	0.8362 (+)
	std	0.0225	0.0222	0.0248	0.0169
leukemia	mean	**0.9764**	0.9130 (+)	0.9095 (+)	0.9189 (+)
	std	0.0215	0.0202	0.0186	0.0090
lung	mean	0.9680	0.9611 (+)	0.9626 (≈)	0.9650 (≈)
	std	0.0086	0.0067	0.0067	0.0043
Avg. Accuracy		**0.9188**	0.8530	0.8610	0.8609

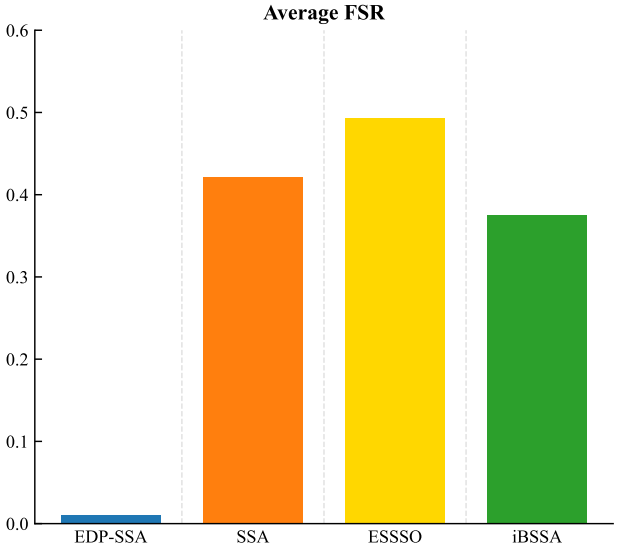


Fig. 9: Average FSR comparison of EDP-SSA and SSA-based algorithms across 16 datasets.

Table 4
Comparison of EDP-SSA and other methods in terms of average and std of accuracy.

Datasets	Stat. measure	EDP-SSA	Other methods						
			BPSO	BGWO	BPSO-V	BinCOA	bIRIME	BMPA-TVSinV	SVM-RFE
11_Tumors	mean	**0.8366**	0.7493(+)	0.7440(+)	0.7425(+)	0.7540(+)	0.7853(+)	0.7509(+)	0.5928(+)
	std	0.0217	0.0120	0.0115	0.0112	0.0134	0.0227	0.0077	0.0000
ALLAML	mean	**0.9629**	0.8823(+)	0.8825(+)	0.8725(+)	0.8825(+)	0.9280(+)	0.8752(+)	0.9161(+)
	std	0.0258	0.0072	0.0074	0.0096	0.0074	0.0257	0.0170	0.0000
AP_Breast Uterus	mean	**0.9752**	0.9600(+)	0.9570(+)	0.9553(+)	0.9579(+)	0.9667(+)	0.9577(+)	0.9273(+)
	std	0.0038	0.0029	0.0026	0.0034	0.0032	0.0053	0.0040	0.0000
AP_Lung Kidney	mean	**0.9848**	0.9599(+)	0.9596(+)	0.9593(+)	0.9591(+)	0.9712(+)	0.9606(+)	0.9250(+)
	std	0.0039	0.0018	0.0024	0.0029	0.0023	0.0062	0.0034	0.0000
CLL_SUB 111	mean	**0.8007**	0.6273(+)	0.6264(+)	0.6001(+)	0.6092(+)	0.6964(+)	0.6282(+)	0.5848(+)
	std	0.0307	0.0212	0.0147	0.0145	0.0209	0.0253	0.0265	0.0000
DLBCL	mean	**0.9613**	0.9416(≈)	0.9439(+)	0.9348(≈)	0.9427(≈)	0.9466(≈)	0.9387(+)	0.9107(+)
	std	0.0239	0.0132	0.0096	0.0090	0.0072	0.0168	0.0123	0.0000
GLI_85	mean	**0.9210**	0.8882(+)	0.8831(+)	0.8832(+)	0.8856(+)	0.9029(+)	0.8893(+)	0.8583(+)
	std	0.0125	0.0066	0.0106	0.0092	0.0085	0.0112	0.0230	0.0000
Leukemia-3	mean	**0.9733**	0.9600(+)	0.9570(+)	0.9553(+)	0.9579(+)	0.9667(+)	0.9577(+)	0.9273(+)
	std	0.0034	0.0029	0.0026	0.0034	0.0032	0.0053	0.0040	0.0000
Prostate_GE	mean	**0.9423**	0.8718(+)	0.8755(+)	0.8736(+)	0.8777(+)	0.9225(+)	0.8771(+)	0.8900(+)
	std	0.0147	0.0109	0.0104	0.0079	0.0067	0.0158	0.0132	0.0000
SMK_CAN 187	mean	**0.7540**	0.6999(+)	0.7002(+)	0.6938(+)	0.7031(+)	0.7157(+)	0.6850(+)	0.7208(+)
	std	0.0220	0.0073	0.0112	0.0096	0.0069	0.0186	0.0185	0.0000
SRBCT	mean	**0.9764**	0.9056(+)	0.8890(+)	0.8738(+)	0.8990(+)	0.9568(≈)	0.9089(+)	0.8194(+)
	std	0.0194	0.0264	0.0123	0.0143	0.0232	0.0277	0.0195	0.0000
TOX_171	mean	**0.8522**	0.7896(+)	0.7837(+)	0.7787(+)	0.7813(+)	0.8134(+)	0.7975(+)	0.7546(+)
	std	0.0255	0.0127	0.0172	0.0120	0.0159	0.0142	0.0151	0.0000
arcene	mean	**0.8825**	0.8680(+)	0.8685(≈)	0.8640(+)	0.8655(+)	0.8720(≈)	0.8425(+)	0.7600(+)
	std	0.0169	0.0075	0.0113	0.0102	0.0098	0.0175	0.0111	0.0000
colon	mean	**0.9336**	0.8533(+)	0.8433(+)	0.8126(+)	0.8407(+)	0.8976(+)	0.8462(+)	0.6929(+)
	std	0.0225	0.0134	0.0208	0.0168	0.0193	0.0285	0.0217	0.0000
leukemia	mean	**0.9764**	0.9284(+)	0.9175(+)	0.9136(+)	0.9218(+)	0.9661(≈)	0.9243(+)	0.8750(+)
	std	0.0215	0.0140	0.0117	0.0098	0.0090	0.0256	0.0245	0.0000
lung	mean	0.9680	0.9660(≈)	0.9674(≈)	0.9660(≈)	0.9689(≈)	**0.9759**(−)	0.9651(≈)	0.9457(+)
	std	0.0086	0.0044	0.0059	0.0043	0.0046	0.0036	0.0104	0.0000
Avg. Accuracy		**0.9188**	0.8657	0.8624	0.8549	0.8629	0.8927	0.8628	0.8188

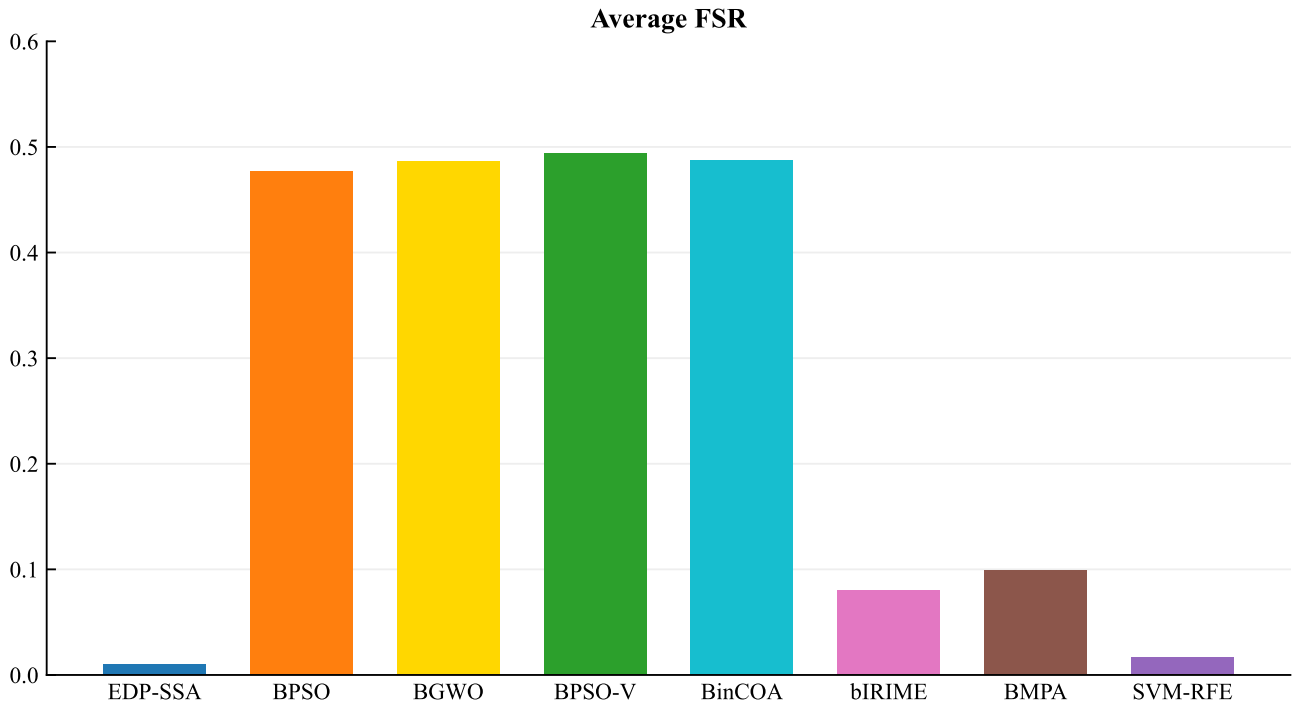


Fig. 10: Average FSR comparison of EDP-SSA and other compared algorithms across 16 datasets.

Table 5
Comparison of EDP-SSA and SSA based methods in terms of average and std of FSR.

Datasets	Stat. measure	EDP-SSA	SSA based		
			SSA Basic	iBSSA	ESSSO
11_Tumors	mean	**0.0117**	0.4223 (+)	0.4237 (+)	0.4997 (+)
	std	0.0010	0.1287	0.0113	0.0089
ALLAML	mean	**0.0084**	0.3825 (+)	0.3777 (+)	0.4897 (+)
	std	0.0009	0.1662	0.0124	0.0117
AP_Breast Uterus	mean	**0.0106**	0.4499 (+)	0.3730 (+)	0.4921 (+)
	std	0.0010	0.0951	0.0211	0.0049
AP_Lung Kidney	mean	**0.0089**	0.4172 (+)	0.3657 (+)	0.4884 (+)
	std	0.0009	0.1101	0.0311	0.0050
CLL_SUB 111	mean	**0.0112**	0.4397 (+)	0.4193 (+)	0.5125 (+)
	std	0.0012	0.1459	0.0369	0.0216
DLBCL	mean	**0.0073**	0.3558 (+)	0.3531 (≈)	0.4909 (≈)
	std	0.0012	0.1861	0.0398	0.0070
GLI_85	mean	**0.0092**	0.4472 (+)	0.4221 (+)	0.4883 (+)
	std	0.0007	0.0732	0.0494	0.0052
Leukemia-3	mean	**0.0107**	0.4499 (+)	0.3730 (+)	0.4921 (+)
	std	0.0012	0.0951	0.0211	0.0049
Prostate_GE	mean	**0.0083**	0.4237 (+)	0.3574 (+)	0.4836 (+)
	std	0.0010	0.1322	0.0319	0.0066
SMK_CAN 187	mean	**0.0107**	0.3991 (+)	0.4304 (+)	0.5017 (+)
	std	0.0009	0.1473	0.0409	0.0090
SRBCT	mean	**0.0104**	0.4271 (+)	0.3091 (+)	0.4883 (+)
	std	0.0026	0.0938	0.0236	0.0163
TOX_171	mean	**0.0143**	0.4314 (+)	0.3925 (+)	0.5009 (+)
	std	0.0021	0.1275	0.0094	0.0106
arcene	mean	**0.0168**	0.4490 (+)	0.4235 (≈)	0.5024 (≈)
	std	0.0190	0.1148	0.0075	0.0048
colon	mean	**0.0090**	0.4486 (+)	0.3114 (+)	0.4955 (+)
	std	0.0025	0.0900	0.0174	0.0235
leukemia	mean	**0.0087**	0.4648 (+)	0.3667 (+)	0.4907 (+)
	std	0.0009	0.0652	0.0400	0.0122
lung	mean	**0.0100**	0.3400 (+)	0.3000 (≈)	0.4768 (≈)
	std	0.0017	0.1522	0.0171	0.0100
Avg. FSR		**0.0104**	0.4218	0.3749	0.4933

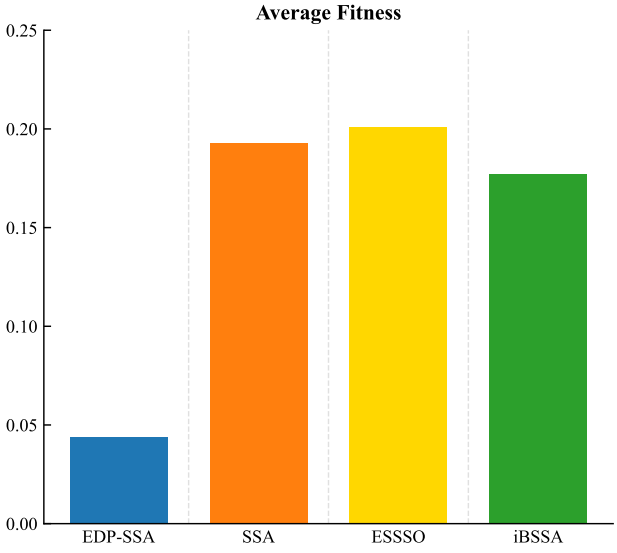


Fig. 11: Average fitness comparison of EDP-SSA and SSA-based algorithms across 16 datasets.

Table 6
Comparison of EDP-SSA and other methods in terms of average and std of FSR.

Datasets	Stat. measure	EDP-SSA	Other methods						
			BPSO	BGWO	BPSO-V	BinCOA	bIRIME	BMPA-TVSinV	SVM-RFE
11_Tumors	mean	**0.0117**	0.4938(+)	0.4933(+)	0.4988(+)	0.4963(+)	0.1196(+)	0.1045(+)	0.0156(+)
	std	0.0010	0.0065	0.0055	0.0042	0.0047	0.0323	0.0540	0.0000
ALLAML	mean	**0.0084**	0.4659(+)	0.4817(+)	0.4940(+)	0.4864(+)	0.0494(+)	0.0985(+)	0.0006(+)
	std	0.0009	0.0086	0.0054	0.0078	0.0037	0.0204	0.0249	0.0000
AP_Breast Uterus	mean	**0.0106**	0.4795(+)	0.4874(+)	0.4916(+)	0.4853(+)	0.0411(+)	0.0385(+)	0.0156(+)
	std	0.0010	0.0079	0.0049	0.0050	0.0052	0.0100	0.0088	0.0000
AP_Lung Kidney	mean	**0.0089**	0.4752(+)	0.4862(+)	0.4930(+)	0.4864(+)	0.0338(+)	0.0348(+)	0.0130(+)
	std	0.0009	0.0114	0.0055	0.0052	0.0035	0.0116	0.0100	0.0000
CLL_SUB 111	mean	**0.0112**	0.4902(+)	0.4962(+)	0.4972(+)	0.4971(+)	0.1240(+)	0.1291(+)	0.0042(+)
	std	0.0012	0.0044	0.0051	0.0066	0.0054	0.0392	0.0503	0.0000
DLBCL	mean	**0.0073**	0.4704(≈)	0.4846(+)	0.4955(≈)	0.4824(≈)	0.0573(≈)	0.1549(+)	0.0005(+)
	std	0.0012	0.0168	0.0060	0.0094	0.0086	0.0202	0.0575	0.0000
GLI_85	mean	**0.0092**	0.4730(+)	0.4845(+)	0.4942(+)	0.4878(+)	0.0624(+)	0.0762(+)	0.0004(+)
	std	0.0007	0.0094	0.0041	0.0046	0.0039	0.0198	0.0253	0.0000
Leukemia-3	mean	**0.0107**	0.4795(+)	0.4874(+)	0.4916(+)	0.4853(+)	0.0411(+)	0.0385(+)	0.0156(+)
	std	0.0012	0.0079	0.0049	0.0050	0.0052	0.0100	0.0088	0.0000
Prostate_GE	mean	**0.0083**	0.4668(+)	0.4816(+)	0.4923(+)	0.4814(+)	0.0828(+)	0.0885(+)	0.0013(+)
	std	0.0010	0.0121	0.0068	0.0066	0.0067	0.0459	0.0329	0.0000
SMK_CAN 187	mean	**0.0107**	0.4912(+)	0.4942(+)	0.4973(+)	0.4941(+)	0.0750(+)	0.0871(+)	0.0004(+)
	std	0.0009	0.0043	0.0038	0.0042	0.0047	0.0304	0.0366	0.0000
SRBCT	mean	**0.0104**	0.4766(+)	0.4818(+)	0.4958(+)	0.4855(+)	0.1114(≈)	0.1509(+)	0.0659(+)
	std	0.0026	0.0077	0.0109	0.0093	0.0112	0.0510	0.0573	0.0000
TOX_171	mean	**0.0143**	0.4847(+)	0.4933(+)	0.4937(+)	0.4950(+)	0.1219(+)	0.1122(+)	0.0157(+)
	std	0.0021	0.0034	0.0062	0.0097	0.0059	0.0360	0.0492	0.0000
arcene	mean	**0.0168**	0.4970(+)	0.4954(≈)	0.5014(+)	0.4947(+)	0.1524(≈)	0.1441(+)	0.0021(+)
	std	0.0190	0.0060	0.0053	0.0049	0.0041	0.0274	0.0527	0.0000
colon	mean	**0.0090**	0.4631(+)	0.4844(+)	0.4851(+)	0.4913(+)	0.1113(+)	0.1442(+)	0.0685(+)
	std	0.0025	0.0102	0.0075	0.0074	0.0137	0.0753	0.0414	0.0000
leukemia	mean	**0.0087**	0.4792(+)	0.4827(+)	0.4918(+)	0.4845(+)	0.0554(≈)	0.1035(+)	0.0003(+)
	std	0.0009	0.0071	0.0038	0.0088	0.0057	0.0338	0.0300	0.0000
lung	mean	**0.0100**	0.4436(≈)	0.4666(≈)	0.4808(≈)	0.4659(≈)	0.0382(−)	0.0703(≈)	0.0471(+)
	std	0.0017	0.0191	0.0095	0.0067	0.0104	0.0115	0.0184	0.0000
Avg. FSR		**0.0104**	0.4769	0.4863	0.4934	0.4875	0.0798	0.0985	0.0167

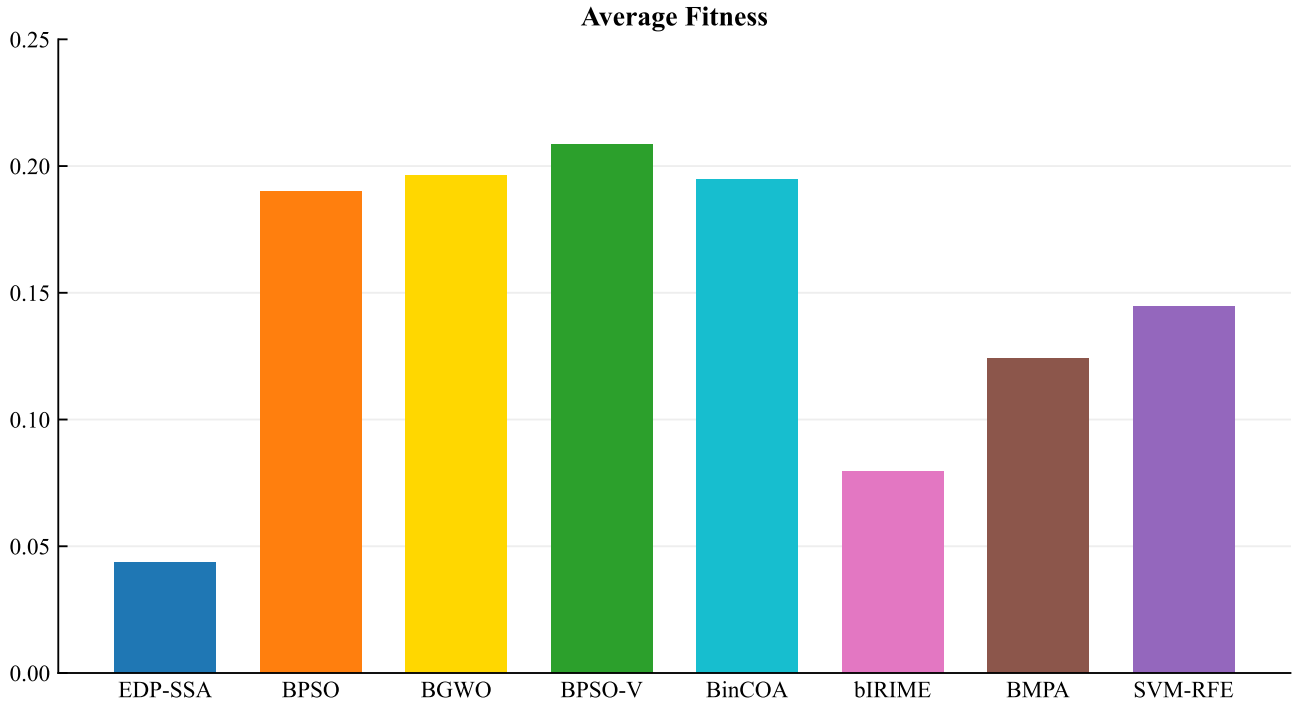


Fig. 12: Average fitness comparison of EDP-SSA and other compared algorithms across 16 datasets.

Table 7
Comparison of EDP-SSA and SSA based methods in terms of average and std of fitness.

Datasets	Stat. measure	EDP-SSA	SSA based		
			SSA Basic	iBSSA	ESSSO
11_Tumors	mean	**0.0989**	0.2739 (+)	0.2687 (+)	0.2847 (+)
	std	0.0153	0.0236	0.0091	0.0061
ALLAML	mean	**0.0168**	0.1828 (+)	0.1806 (+)	0.1965 (+)
	std	0.0116	0.0196	0.0080	0.0066
AP_Breast Uterus	mean	**0.0173**	0.1235 (+)	0.1110 (+)	0.1309 (+)
	std	0.0030	0.0160	0.0025	0.0012
AP_Lung Kidney	mean	**0.0092**	0.1163 (+)	0.1070 (+)	0.1291 (+)
	std	0.0020	0.0205	0.0057	0.0014
CLL_SUB 111	mean	**0.1092**	0.3855 (+)	0.3693 (+)	0.3849 (+)
	std	0.0230	0.0170	0.0086	0.0137
DLBCL	mean	**0.0046**	0.1214 (+)	0.1165 (≈)	0.1380 (≈)
	std	0.0050	0.0169	0.0056	0.0057
GLI_85	mean	**0.0414**	0.1930 (+)	0.1879 (+)	0.1955 (+)
	std	0.0107	0.0106	0.0055	0.0041
Leukemia-3	mean	**0.0189**	0.1235 (+)	0.1110 (+)	0.1309 (+)
	std	0.0023	0.0160	0.0025	0.0012
Prostate_GE	mean	**0.0296**	0.1747 (+)	0.1583 (+)	0.1804 (+)
	std	0.0066	0.0141	0.0042	0.0034
SMK_CAN 187	mean	**0.1567**	0.3227 (+)	0.3226 (+)	0.3299 (+)
	std	0.0082	0.0192	0.0082	0.0043
SRBCT	mean	**0.0030**	0.2060 (+)	0.1523 (+)	0.2093 (+)
	std	0.0032	0.0142	0.0190	0.0141
TOX_171	mean	**0.0739**	0.2291 (+)	0.2124 (+)	0.2374 (+)
	std	0.0173	0.0156	0.0088	0.0043
arcene	mean	**0.0642**	0.1878 (+)	0.1711 (≈)	0.1961 (≈)
	std	0.0092	0.0180	0.0056	0.0055
colon	mean	**0.0309**	0.1904 (+)	0.1394 (+)	0.1907 (+)
	std	0.0102	0.0132	0.0128	0.0113
leukemia	mean	**0.0072**	0.1583 (+)	0.1388 (+)	0.1592 (+)
	std	0.0058	0.0098	0.0112	0.0062
lung	mean	**0.0166**	0.0932 (+)	0.0840 (≈)	0.1174 (≈)
	std	0.0051	0.0278	0.0022	0.0019
Avg. Fitness		**0.0436**	0.1926	0.1769	0.2007

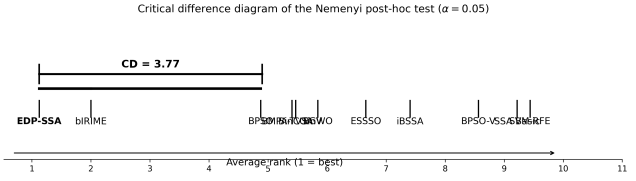


Fig. 13: Critical difference diagram of the Nemenyi post-hoc test ($\alpha = 0.05$) over the average ranks of the 11 algorithms across the 16 datasets.

Table 8
Comparison of EDP-SSA and other methods in terms of average and std of fitness.

Datasets	Stat. measure	EDP-SSA	Other methods						
			BPSO	BGWO	BPSO-V	BinCOA	bIRIME	BMPA-TVSinV	SVM-RFE
11_Tumors	mean	**0.0989**	0.2734(+)	0.2830(+)	0.2961(+)	0.2759(+)	0.1554(+)	0.2062(+)	0.3569(+)
	std	0.0153	0.0088	0.0048	0.0031	0.0077	0.0218	0.0151	0.0000
ALLAML	mean	**0.0168**	0.1829(+)	0.1892(+)	0.2026(+)	0.1870(+)	0.0494(+)	0.1118(+)	0.0656(+)
	std	0.0116	0.0083	0.0066	0.0051	0.0080	0.0234	0.0084	0.0000
AP_Breast Uterus	mean	**0.0173**	0.1267(+)	0.1288(+)	0.1318(+)	0.1283(+)	0.0299(+)	0.0393(+)	0.0578(+)
	std	0.0030	0.0028	0.0019	0.0006	0.0013	0.0046	0.0035	0.0000
AP_Lung Kidney	mean	**0.0092**	0.1254(+)	0.1274(+)	0.1306(+)	0.1266(+)	0.0256(+)	0.0369(+)	0.0688(+)
	std	0.0020	0.0024	0.0017	0.0007	0.0013	0.0052	0.0035	0.0000
CLL_SUB 111	mean	**0.1092**	0.3688(+)	0.3817(+)	0.4072(+)	0.3806(+)	0.2256(+)	0.3178(+)	0.2876(+)
	std	0.0230	0.0176	0.0096	0.0068	0.0079	0.0249	0.0160	0.0000
DLBCL	mean	**0.0046**	0.1298(≈)	0.1337(+)	0.1439(≈)	0.1302(≈)	0.0236(≈)	0.0719(+)	0.0841(+)
	std	0.0050	0.0056	0.0075	0.0062	0.0039	0.0099	0.0101	0.0000
GLI_85	mean	**0.0414**	0.1934(+)	0.1938(+)	0.1977(+)	0.1917(+)	0.0633(+)	0.1103(+)	0.1130(+)
	std	0.0107	0.0076	0.0057	0.0041	0.0008	0.0067	0.0104	0.0000
Leukemia-3	mean	**0.0189**	0.1267(+)	0.1288(+)	0.1318(+)	0.1283(+)	0.0299(+)	0.0393(+)	0.0578(+)
	std	0.0023	0.0028	0.0019	0.0006	0.0013	0.0046	0.0035	0.0000
Prostate_GE	mean	**0.0296**	0.1747(+)	0.1777(+)	0.1845(+)	0.1767(+)	0.0633(+)	0.1023(+)	0.0784(+)
	std	0.0066	0.0071	0.0041	0.0025	0.0045	0.0179	0.0058	0.0000
SMK_CAN 187	mean	**0.1567**	0.3257(+)	0.3263(+)	0.3347(+)	0.3254(+)	0.2159(+)	0.2535(+)	0.2230(+)
	std	0.0082	0.0052	0.0044	0.0026	0.0022	0.0170	0.0091	0.0000
SRBCT	mean	**0.0030**	0.1796(+)	0.1928(+)	0.2255(+)	0.1923(+)	0.0380(≈)	0.1299(+)	0.1308(+)
	std	0.0032	0.0206	0.0127	0.0042	0.0138	0.0275	0.0192	0.0000
TOX_171	mean	**0.0739**	0.2211(+)	0.2326(+)	0.2453(+)	0.2334(+)	0.1302(+)	0.1760(+)	0.2231(+)
	std	0.0173	0.0093	0.0058	0.0033	0.0051	0.0112	0.0082	0.0000
arcene	mean	**0.0642**	0.1930(+)	0.1955(≈)	0.2035(+)	0.1961(+)	0.1121(≈)	0.1436(+)	0.1884(+)
	std	0.0092	0.0080	0.0050	0.0035	0.0024	0.0116	0.0075	0.0000
colon	mean	**0.0309**	0.1630(+)	0.1847(+)	0.2149(+)	0.1797(+)	0.0651(+)	0.1347(+)	0.2342(+)
	std	0.0102	0.0118	0.0137	0.0097	0.0085	0.0227	0.0109	0.0000
leukemia	mean	**0.0072**	0.1460(+)	0.1541(+)	0.1691(+)	0.1526(+)	0.0221(≈)	0.0737(+)	0.1006(+)
	std	0.0058	0.0072	0.0101	0.0047	0.0034	0.0134	0.0158	0.0000
lung	mean	**0.0166**	0.1120(≈)	0.1130(≈)	0.1190(≈)	0.1125(≈)	0.0206(−)	0.0361(≈)	0.0410(+)
	std	0.0051	0.0048	0.0025	0.0010	0.0022	0.0037	0.0026	0.0000
Avg. Fitness		**0.0436**	0.1901	0.1964	0.2086	0.1948	0.0794	0.1240	0.1445

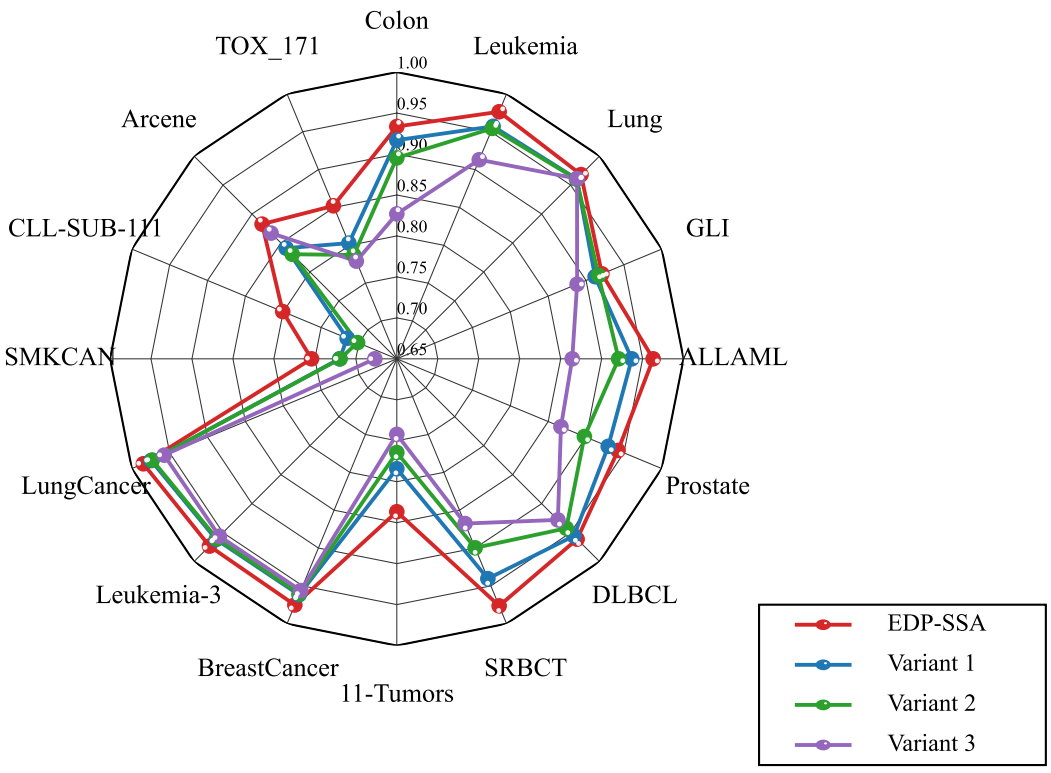
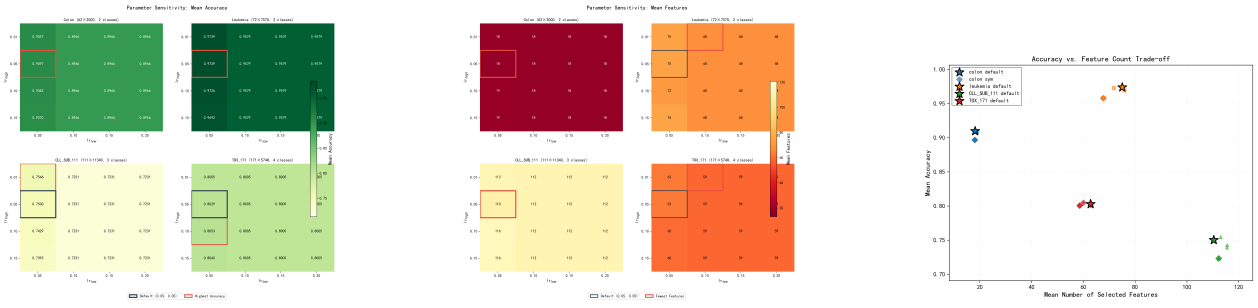


Fig. 14: Radar chart comparing the average accuracy of EDP-SSA and its ablation variants across the 16 datasets.



(a) Accuracy (b) Feature count (c) Accuracy vs. feature count

Fig. 15: EDP-SSA parameter sensitivity analysis.

Table 9
Comparison of EDP-SSA and SSA based methods in terms of average and std of runtime.

Datasets	Stat. measure	EDP-SSA	SSA based		
			SSA Basic	iBSSA	ESSSO
11_Tumors	mean	482.35	557.26	326.08	**199.81**
	std	9.36	2.15	15.49	33.50
ALLAML	mean	651.59	561.81	236.11	**150.66**
	std	0.66	6.99	10.50	30.59
AP_Breast Uterus	mean	485.92	811.04	455.99	**318.34**
	std	3.02	32.31	12.11	54.61
AP_Lung Kidney	mean	428.96	664.80	413.03	**278.88**
	std	39.66	11.48	12.32	46.34
CLL_SUB 111	mean	441.52	425.33	284.39	**171.18**
	std	9.30	2.05	11.31	31.37
DLBCL	mean	473.20	414.78	226.95	**149.34**
	std	1.14	2.59	8.75	32.49
GLI_85	mean	654.73	659.93	380.95	**180.78**
	std	0.92	7.42	8.52	25.44
Leukemia-3	mean	427.52	751.52	460.54	**317.92**
	std	45.44	26.43	9.95	55.15
Prostate_GE	mean	473.96	428.23	233.03	**153.23**
	std	1.48	1.55	11.30	32.82
SMK_CAN 187	mean	430.01	620.63	454.94	**256.26**
	std	39.75	5.28	18.07	34.57
SRBCT	mean	424.73	2654.72	216.49	**145.61**
	std	93.27	1942.41	0.68	37.05
TOX_171	mean	440.31	405.23	244.89	**162.30**
	std	11.45	4.94	8.45	35.62
arcene	mean	438.31	468.89	305.63	**193.70**
	std	17.81	3.48	9.38	36.71
colon	mean	491.87	436.22	213.60	**143.53**
	std	27.32	2.92	0.92	37.36
leukemia	mean	531.91	457.53	243.47	**149.96**
	std	5.67	3.05	6.03	30.97
lung	mean	440.05	381.61	228.98	**153.82**
	std	8.30	4.68	3.26	37.64

Table 10
Comparison of EDP-SSA and other methods in terms of average and std of runtime.

Datasets	Stat. measure	EDP-SSA	Other methods						
			BPSO	BGWO	BPSO-V	BinCOA	bIRIME	BMPA-TVSinV	SVM-RFE
11_Tumors	mean	482.35	282.70	285.77	276.37	565.08	672.87	445.92	**19.38**
	std	9.36	0.83	1.99	2.85	9.01	14.08	105.43	1.34
ALLAML	mean	651.59	281.05	281.01	208.60	419.54	597.10	349.06	**0.93**
	std	0.66	0.68	1.21	1.49	2.84	11.47	75.82	0.01
AP_Breast Uterus	mean	485.92	495.51	504.30	497.66	895.42	568.84	800.25	**23.73**
	std	3.02	8.65	2.07	6.14	136.05	15.86	47.82	2.08
AP_Lung Kidney	mean	428.96	335.11	344.14	337.20	682.95	557.37	693.78	**14.43**
	std	39.66	2.90	1.80	5.12	15.26	13.05	38.07	0.14
CLL_SUB 111	mean	441.52	210.67	212.20	196.09	420.88	561.94	379.11	**5.52**
	std	9.30	2.01	1.02	6.03	1.39	13.97	53.07	0.04
DLBCL	mean	473.20	207.60	207.88	204.09	415.53	595.61	368.93	**0.90**
	std	1.14	1.60	1.67	2.01	2.79	9.83	40.21	0.01
GLI_85	mean	654.73	328.56	330.53	303.49	657.43	841.36	519.53	**5.19**
	std	0.92	2.61	1.10	8.28	2.85	7.27	121.98	0.12
Leukemia-3	mean	427.52	374.90	388.24	394.71	786.13	573.94	779.22	**24.05**
	std	45.44	5.93	1.71	4.02	30.05	19.14	62.35	0.16
Prostate_GE	mean	473.96	215.17	215.49	212.10	420.74	604.25	387.21	**0.99**
	std	1.48	1.44	1.25	1.82	13.23	6.05	35.40	0.03
SMK_CAN 187	mean	430.01	306.47	294.19	275.05	624.18	569.32	653.41	**28.68**
	std	39.75	9.83	3.50	5.04	8.57	23.68	33.08	0.33
SRBCT	mean	424.73	198.34	196.58	196.06	393.31	564.03	340.75	**4.16**
	std	93.27	3.58	0.62	1.37	1.54	81.72	45.23	0.92
TOX_171	mean	440.31	198.91	198.67	190.34	387.56	552.69	356.79	**2.85**
	std	11.45	1.77	0.71	2.97	5.85	4.81	31.16	0.03
arcene	mean	438.31	234.22	235.39	221.67	464.66	576.23	451.88	**4.13**
	std	17.81	2.16	2.03	6.23	1.28	15.75	36.51	0.10
colon	mean	491.87	217.61	217.80	215.85	435.27	657.25	410.48	**2.43**
	std	27.32	1.13	0.68	1.11	0.92	8.25	22.99	0.08
leukemia	mean	531.91	228.37	229.46	221.62	424.29	544.84	339.34	**1.19**
	std	5.67	1.29	1.23	1.08	26.99	7.71	40.64	0.03
lung	mean	440.05	283.72	284.21	280.48	566.87	689.42	419.02	**4.36**
	std	8.30	0.81	1.22	1.18	1.97	207.45	95.33	1.13

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The datasets used in this study are publicly available microarray datasets, as summarized in Table 1. The source references are provided in the dataset descriptions and bibliography.

CRediT authorship contribution statement

Ying Zhang: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Writing – Original Draft, Visualization.